

Lecture 13

Learning Quantum States and Hamiltonians

Deep Learning in Classical and Quantum Physics

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Motivation



Certifying a Source of Quantum States

A device is claimed to prepare a quantum state $\hat{\rho}$, for example an entangled state of a few qubits. It is a black box that returns one copy per run. We want to know

- whether it prepares the intended state,
- whether the state is pure,
- whether it is entangled, and how many qubits share the entanglement.

All answers follow from the density matrix. **Quantum state tomography** reconstructs it from measurements on many copies. Every copy costs time, so the number of copies is limited and the reconstruction is noisy.

We reconstruct a state by the standard methods, linear inversion and maximum likelihood, and then train a neural network that removes the noise of such a reconstruction. The second part turns from the state to the Hamiltonian that generates its dynamics.

A Device Returns Bit Strings

A quantum device prepares a state $\hat{\rho}$ of N qubits, rotates every qubit with a single-qubit gate and measures all qubits in the computational basis. One run destroys the state and returns one bit string $\mathbf{b} = b_0 b_1 \ldots b_{N-1}$. The choice of rotations is called a **measurement setting** s, and the Born rule gives the probability of each outcome,

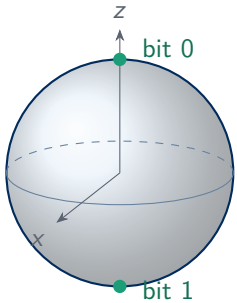
$$p_s(\mathbf{b}) = \mathrm{Tr}\big[\hat{\rho}\,\hat{\pi}_{s,\mathbf{b}}\big], \qquad \hat{\pi}_{s,\mathbf{b}} = \underbrace{\hat{U}_s^\dagger |\mathbf{b}\rangle\langle\mathbf{b}| \hat{U}_s}_{\text{projector of outcome } \mathbf{b} \text{ in setting } s} .$$

Repeating the preparation M times per setting gives counts $n_s(\mathbf{b})$ and frequencies $f_s(\mathbf{b}) = n_s(\mathbf{b})/M$, which approach $p_s(\mathbf{b})$ with a statistical error proportional to $M^{-1/2}$.

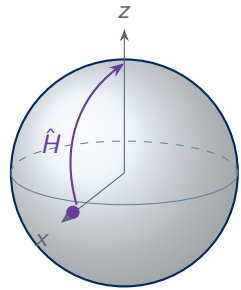
Conventions

$|0\rangle$ has $\hat{Z}|0\rangle = +|0\rangle$, and bit 0 signals the eigenvalue $+1$ of the measured Pauli operator. Qubit 0 is the most significant bit. To measure $\hat{X}$ we apply the Hadamard gate $\hat{H}$, to measure $\hat{Y}$ we apply $\hat{S}^\dagger$ and then $\hat{H}$, with $\hat{S} = \mathrm{diag}(1, i)$.

Choosing the Measured Operator



measure $\hat{Z}$ directly



measure $\hat{X}$ by applying $\hat{H}$ before $\hat{Z}$

$\hat{H}$ swaps the x and z axes, so bit 0 signals $\hat{X} = +1$ (for $\hat{Y}$ apply $\hat{S}^\dagger$ first)

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Quantum State Tomography

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Density Matrices

A device that prepares $|\psi_a\rangle$ with probability w_a is described by the density matrix $\hat{\rho} = \sum_a w_a |\psi_a\rangle\langle\psi_a|$. A physical state is Hermitian, has unit trace and non-negative eigenvalues, $\hat{\rho} \geq 0$. Its purity $\mathrm{Tr}\hat{\rho}^2$ is one for a pure state and $1/d$ for the maximally mixed state $\hat{I}/d$, with $d = 2^N$.

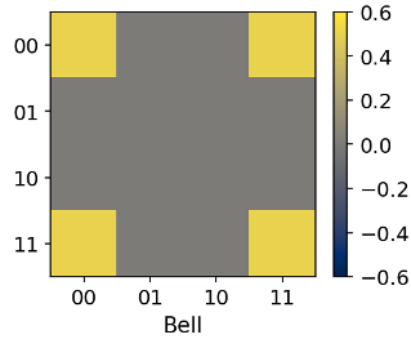
Our three test states of two qubits are

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \qquad \hat{\rho}_W = v\,|\Phi^+\rangle\langle\Phi^+| + (1-v)\frac{\hat{I}}{4}, \qquad \hat{\rho}_{\mathrm{rand}} = \frac{GG^\dagger}{\mathrm{Tr}(GG^\dagger)}.$$

The Bell state is pure and entangled. The Werner state with visibility $v = 0.8$ is a Bell state mixed with white noise and is entangled for $v > 1/3$. The random state, with a 4×4 matrix G of independent complex Gaussian entries, samples the Hilbert–Schmidt measure.

R. F. Werner, PRA **40**, 4277 (1989), doi:10.1103/PhysRevA.40.4277 A. Peres, PRL **77**, 1413 (1996), doi:10.1103/PhysRevLett.77.1413 M. Horodecki *et al.*, Phys. Lett. A **223**, 1 (1996), doi:10.1016/S0375-9601(96)00706-2

The Bell State as a Matrix

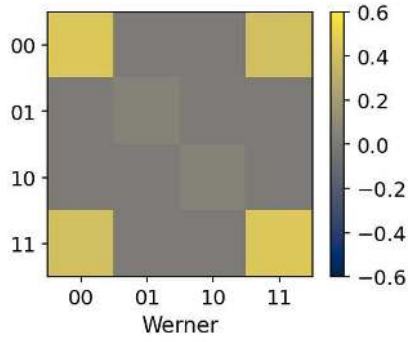


our simulation

Real parts $\mathrm{Re}\,\rho_{ij}$ of the Bell state in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Populations of $|00\rangle$ and $|11\rangle$ and the coherence between them fill the four corners.

The Werner State as a Matrix

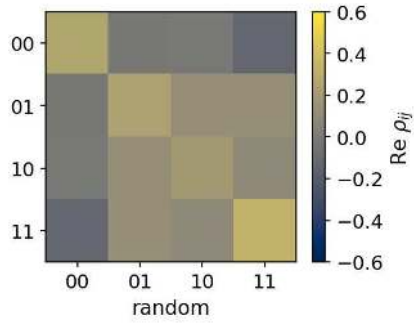


our simulation

Real parts of the Werner state, $v = 0.8$.

The same pattern with reduced coherence and a uniform background on the diagonal.

A Random Mixed State



our simulation

Real parts of the Hilbert–Schmidt random state.

Nonzero entries everywhere, and an imaginary part not shown. Tomography has to recover such matrices from bit strings alone.

K. Życzkowski and H.-J. Sommers, J. Phys. A **34**, 7111 (2001), doi:10.1088/0305-4470/34/35/335

Distances Between States

An estimate $\tilde{\rho}$ is judged by its distance to the true state $\hat{\rho}$, known in a simulation.

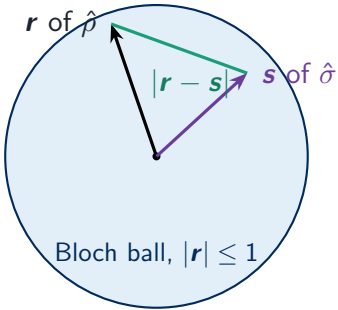
$$F(\hat{\rho},\tilde{\rho}) = \Big(\mathrm{Tr}\sqrt{\sqrt{\hat{\rho}}\,\tilde{\rho}\,\sqrt{\hat{\rho}}}\Big)^2, \qquad D = \tfrac{1}{2}\|\hat{\rho}-\tilde{\rho}\|_1, \qquad D_{\mathrm{HS}} = \sqrt{\mathrm{Tr}[(\hat{\rho}-\tilde{\rho})^2]}.$$

- The fidelity F is one for identical states and $\langle\psi|\tilde{\rho}|\psi\rangle$ for a pure $\hat{\rho} = |\psi\rangle\langle\psi|$. It needs a physical $\tilde{\rho}$.
- The trace distance uses the trace norm $\|\hat{A}\|_1 = \mathrm{Tr}\sqrt{\hat{A}^\dagger\hat{A}}$, the sum of the moduli of the eigenvalues of a Hermitian $\hat{A}$. D is the largest difference of outcome probabilities.
- D_{HS} is the Euclidean distance of the matrices viewed as vectors.

Bell against Werner gives $F = v + (1-v)/4$ and $D = \frac{3}{4}(1-v)$.

A. Uhlmann, Rep. Math. Phys. **9**, 273 (1976), doi:10.1016/0034-4877(76)90060-4 R. Jozsa, J. Mod. Opt. **41**, 2315 (1994), doi:10.1080/09500349414552171

Distances for One Qubit



trace distance $D = \frac{1}{2}|\boldsymbol{r} - \boldsymbol{s}|$

Hilbert–Schmidt distance

$D_{\mathrm{HS}} = \frac{1}{\sqrt{2}}|\boldsymbol{r} - \boldsymbol{s}|$

fidelity

$$F = \frac{1}{2}\left(1 + \boldsymbol{r}\cdot\boldsymbol{s} + \sqrt{(1-|\boldsymbol{r}|^2)(1-|\boldsymbol{s}|^2)}\right)$$

$F = 1$ for identical states

one qubit, $\hat{\rho} = \frac{1}{2}(\hat{I} + \boldsymbol{r}\cdot\hat{\boldsymbol{\sigma}})$, $\hat{\sigma} = \frac{1}{2}(\hat{I} + \boldsymbol{s}\cdot\hat{\boldsymbol{\sigma}})$

For one qubit the trace distance is half the length of the chord between the two Bloch vectors.

One Qubit from Three Settings

The matrices $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ form an orthogonal basis, $\mathrm{Tr}(\hat{P}\hat{P}') = 2\delta_{PP'}$, so every one-qubit state is

$$\hat{\rho} = \tfrac{1}{2}(\hat{I} + r_x\hat{X} + r_y\hat{Y} + r_z\hat{Z}), \qquad r_P = \langle\hat{P}\rangle = \mathrm{Tr}(\hat{\rho}\hat{P}), \qquad |\boldsymbol{r}| \le 1.$$

The eigenvalues are $(1 \pm |\boldsymbol{r}|)/2$, so physical states fill the Bloch ball and pure states lie on its surface. In setting P the bit 1 has probability $(1 - r_P)/2$. Each shot gives an independent value ± 1 with mean r_P, and n_1 outcomes 1 among M shots give

$$\tilde{r}_P = \underbrace{1 - \frac{2n_1}{M}}_{\text{mean of the } \pm 1 \text{ values}}, \qquad \mathrm{Var}(\tilde{r}_P) = \frac{\overbrace{\langle(\pm 1)^2\rangle}^{=1} - r_P^2}{M} = \frac{1 - r_P^2}{M}.$$

Three settings with M shots each determine the state. The estimate is unbiased, but nothing keeps $|\tilde{\boldsymbol{r}}|$ below one.

The Bloch Vector from Counts

$$\hat{\rho} \xrightarrow{\hat{H}} \tilde{r}_x = \frac{n_0 - n_1}{M} \approx \langle \hat{X} \rangle$$

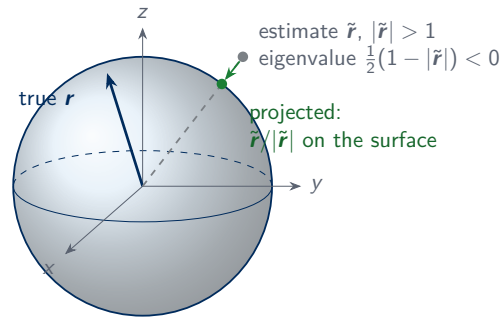
$$\hat{\rho} \xrightarrow{\hat{H}\hat{S}^\dagger} \tilde{r}_y = \frac{n_0 - n_1}{M} \approx \langle \hat{Y} \rangle$$

$$\hat{\rho} \xrightarrow{\hat{I}} \tilde{r}_z = \frac{n_0 - n_1}{M} \approx \langle \hat{Z} \rangle$$

$\hat{H}\hat{S}^\dagger$ applies $\hat{S}^\dagger$, then $\hat{H}$

n_0 and n_1 count the bits 0 and 1 in M shots

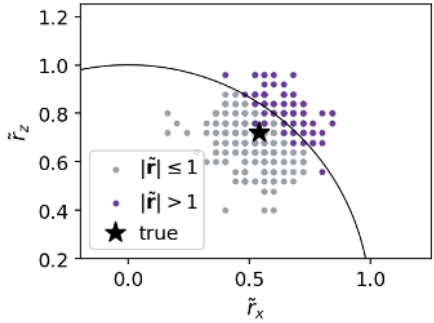
$$\tilde{\rho} = \tfrac{1}{2}(\hat{I} + \tilde{r}_x\hat{X} + \tilde{r}_y\hat{Y} + \tilde{r}_z\hat{Z})$$



physical states have $|\boldsymbol{r}| \leq 1$

An estimate outside the unit ball has the negative eigenvalue $(1 - |\tilde{\boldsymbol{r}}|)/2$. The closest physical state is the unit vector in the same direction.

Estimates Scatter Around the Bloch Vector

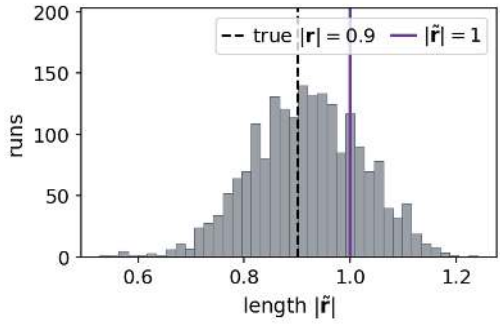


our simulation

Mixed state with $\boldsymbol{r} = (0.54, 0, 0.72)$, $|\boldsymbol{r}| = 0.9$, $M = 50$ shots per setting. Estimates $(\tilde{r}_x, \tilde{r}_z)$, the star marks the true vector and the arc the unit circle.

The estimates scatter around the true vector with the spread of $\mathrm{Var}(\tilde{r}_P)$.

Estimates Leave the Bloch Ball



our simulation

Length $|\tilde{\boldsymbol{r}}|$ of 2000 estimates, the dashed line marks the true length.

23 % of the estimates have $|\tilde{\boldsymbol{r}}| > 1$ and are unphysical.

Pauli Strings and Informational Completeness

The 4^N Pauli strings $\hat{P} = \hat{\sigma}_{p_0} \otimes \cdots \otimes \hat{\sigma}_{p_{N-1}}$, $p_q \in \{I, X, Y, Z\}$, obey $\mathrm{Tr}(\hat{P}\hat{P}') = 2^N \delta_{PP'}$, so

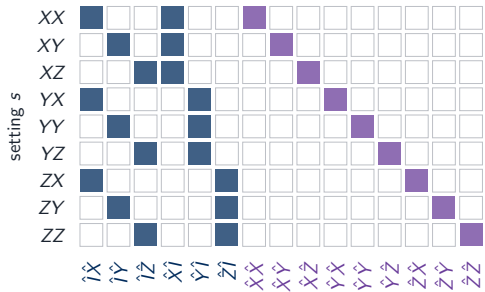
$$\hat{\rho} = \frac{1}{2^N} \sum_P \langle \hat{P} \rangle \, \hat{P}.$$

The identity coefficient is one, so $4^N - 1$ real numbers have to be measured. The number of non-identity letters of $\hat{P}$ is its **weight** k.

- A setting $s = (s_0, \ldots, s_{N-1})$, $s_q \in \{X, Y, Z\}$, rotates every qubit and measures all of them. There are 3^N settings.
- A setting measures every string that agrees with s on its support, since the product of the ± 1 values there is the eigenvalue of the string. A string of weight k is measured by 3^{N-k} settings, $\hat{X}\hat{I}$ by XX, XY and XZ.
- The settings fix all 4^N expectation values, and the measurement is **informationally complete**.

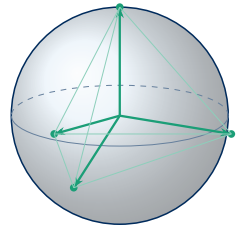
Settings and Pauli Strings

(a) which Pauli strings each setting measures, $N = 2$



weight 1, measured by $3^{N-1} = 3$ settings
weight 2, measured by $3^{N-2} = 1$ setting

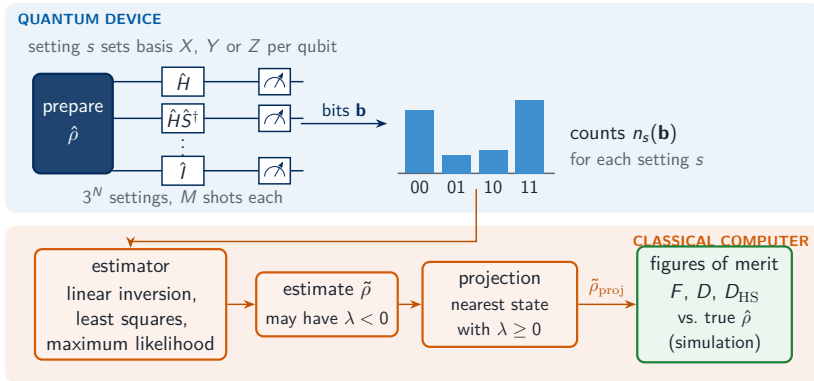
(b) one-qubit SIC measurement



four outcomes $\hat{\Pi}_a = \frac{1}{4}(\hat{I} + \boldsymbol{n}_a \cdot \hat{\boldsymbol{\sigma}})$
$\boldsymbol{n}_a$ at the corners of a tetrahedron

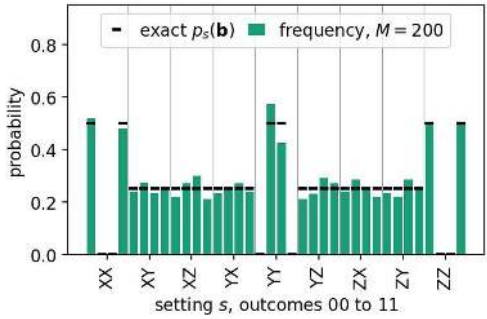
A single informationally complete measurement also exists, for one qubit with four outcomes pointing to the corners of a tetrahedron.

The Tomography Protocol



Initial state the unknown $\hat{\rho}$, prepared on demand. **Operations** single-qubit rotations and measurements in the computational basis, M shots in each of the 3^N settings. **Goal** an estimate $\tilde{\rho}$. **Success** in simulation, a small D or $1 - F$ to the true state.

Counts of the Bell State



our simulation

Frequencies of the four outcomes $00, 01, 10, 11$ in each of the nine settings for the Bell state with $M = 200$ (bars), with the exact probabilities (dashes).

Settings XX and ZZ give only even parities and YY only odd ones, which reflects $\langle \hat{X}\hat{X} \rangle = \langle \hat{Z}\hat{Z} \rangle = 1$ and $\langle \hat{Y}\hat{Y} \rangle = -1$.

Linear Inversion

The Born rule is linear in the Pauli coefficients $r_P = \langle \hat{P} \rangle$,

$$p_s(\mathbf{b}) = \mathrm{Tr}\Big[\hat{\pi}_{s,\mathbf{b}} \, \frac{1}{2^N} \sum_P r_P \hat{P}\Big] = \sum_P \underbrace{\frac{\mathrm{Tr}\big[\hat{\pi}_{s,\mathbf{b}} \hat{P}\big]}{2^N}}_{A_{(s,\mathbf{b}),P}} \, r_P .$$

The design matrix A has 36 rows (setting and outcome) and 16 columns (strings) for two qubits. With frequencies $\boldsymbol{f}$ in place of probabilities,

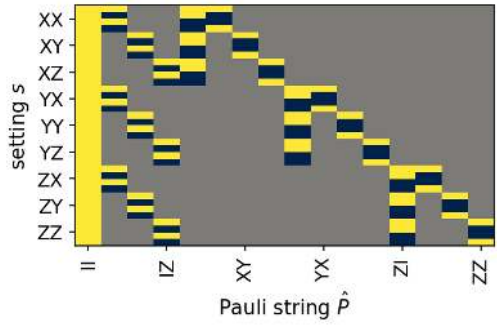
$$\tilde{\boldsymbol{r}} = \arg\min_{\boldsymbol{r}} \|A\boldsymbol{r} - \boldsymbol{f}\|^2 = (A^{\mathsf{T}}A)^{-1} A^{\mathsf{T}} \boldsymbol{f}$$

defines $\tilde{\rho}_{\mathrm{LI}}$. For Pauli settings $A^{\mathsf{T}}A$ is diagonal, and $\tilde{r}_P$ is an average of parities,

$$\tilde{r}_P = \frac{1}{3^{N-k}} \sum_{s \text{ measures } P} \sum_{\mathbf{b}} \underbrace{(-1)^{\sum_{q \in \mathrm{supp} P} b_q}}_{\text{eigenvalue of } \hat{P}} f_s(\mathbf{b}), \qquad \mathrm{Var}(\tilde{r}_P) = \frac{1 - r_P^2}{\underbrace{3^{N-k} M}_{\text{shots that see } \hat{P}}}.$$

D. F. V. James *et al.*, PRA **64**, 052312 (2001), doi:10.1103/PhysRevA.64.052312

The Design Matrix

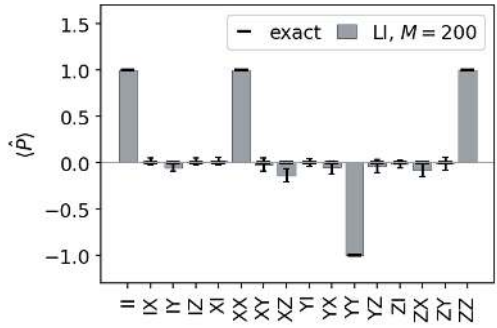


our simulation

$2^N A$ for two qubits, entries $+1$ (yellow), -1 (dark) and 0 (grey).

Weight-one columns have entries in three settings, weight-two columns in one.

Pauli Coefficients of the Bell State

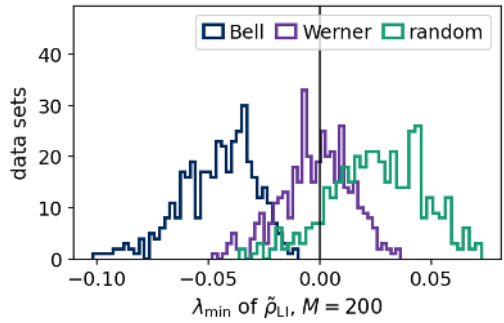


our simulation

Pauli coefficients from $M = 200$ shots per setting with the error bars of $\mathrm{Var}(\tilde{r}_P)$, and the exact values.

Strings with $r_P = \pm 1$ are recovered without error, and weight-one strings fluctuate less than weight-two strings.

Unphysical Estimates

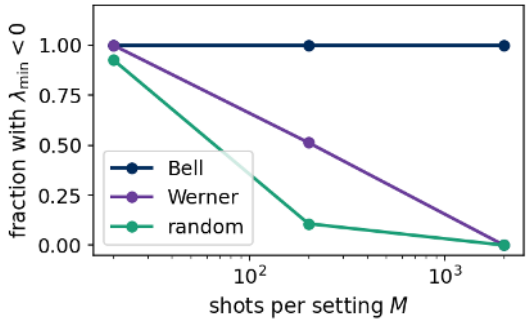


our simulation

Smallest eigenvalue $\lambda_{\min}$ of $\tilde{\rho}_{\mathrm{LI}}$ for 400 data sets at $M = 200$.

For the Bell state every estimate has $\lambda_{\min} < 0$.

Unphysical Estimates at Every Budget



our simulation

Fraction of data sets with $\lambda_{\min} < 0$ against M.

For the pure Bell state the fraction stays at one, since its zero eigenvalues are surrounded by noise of both signs. For full-rank states it falls once the noise is smaller than the smallest eigenvalue.

The Bell Estimate Is Always Unphysical

The settings XX, YY and ZZ return the same parity in every shot, so $\tilde{r}_{XX} = \tilde{r}_{ZZ} = 1$ and $\tilde{r}_{YY} = -1$ without error. The weight of the estimate on the Bell state is therefore exact,

$$\langle \Phi^+ | \tilde{\rho}_{\mathrm{LI}} | \Phi^+ \rangle = \tfrac{1}{4}(1 + \tilde{r}_{XX} - \tilde{r}_{YY} + \tilde{r}_{ZZ}) = 1.$$

With $\mathrm{Tr}\tilde{\rho}_{\mathrm{LI}} = 1$, the restriction $\hat{Q}\tilde{\rho}_{\mathrm{LI}}\hat{Q}$ to the three states orthogonal to $|\Phi^+\rangle$, with the projector $\hat{Q} = \hat{I} - |\Phi^+\rangle\langle\Phi^+|$, has

$$\mathrm{Tr}(\hat{Q}\tilde{\rho}_{\mathrm{LI}}\hat{Q}) = \underbrace{\mathrm{Tr}\tilde{\rho}_{\mathrm{LI}}}_{=1} - \underbrace{\langle \Phi^+ | \tilde{\rho}_{\mathrm{LI}} | \Phi^+ \rangle}_{=1} = 0.$$

A nonzero Hermitian matrix with zero trace has a negative eigenvalue, and the smallest eigenvalue of $\tilde{\rho}_{\mathrm{LI}}$ is at most that of its restriction. Any noise in the other twelve coefficients thus makes the estimate unphysical.

Projection onto the Physical States

The closest physical state in the Hilbert–Schmidt distance,

$$\tilde{\rho}_{\mathrm{proj}} = \arg\min_{\hat{\sigma}\geq 0,\ \mathrm{Tr}\hat{\sigma}=1} \left\| \tilde{\rho}_{\mathrm{LI}} - \hat{\sigma} \right\|_{\mathrm{HS}},$$

keeps the eigenvectors of $\tilde{\rho}_{\mathrm{LI}}$ and changes only its eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d$,

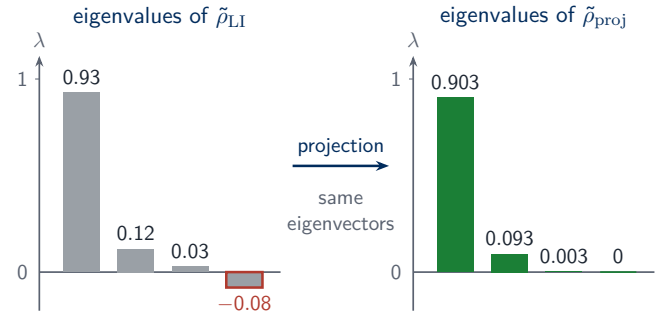
$$\mu_i = \max(\lambda_i - \tau,\, 0), \qquad \underbrace{\textstyle\sum_i \mu_i = 1}_{\text{fixes the shift } \tau}\ .$$

The algorithm walks up from the smallest eigenvalue, sets negative eigenvalues to zero, subtracts the removed weight equally from the others and stops when none is negative. It costs one diagonalisation of a $2^N \times 2^N$ matrix.

The set of physical states is convex and contains the true state, so the projection never increases the Hilbert–Schmidt distance to it.

J. A. Smolin, J. M. Gambetta and G. Smith, PRL **108**, 070502 (2012), doi:10.1103/PhysRevLett.108.070502

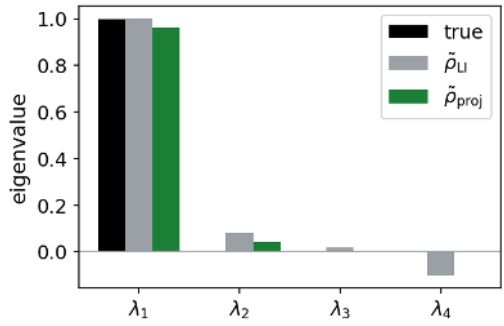
Projecting a Spectrum



the negative eigenvalue is set to 0, and its weight is subtracted equally from the others (repeated while needed), so the trace stays 1

Clipping the negative eigenvalue and rescaling the rest also gives a physical state, but a different one, and it is farther from $\tilde{\rho}_{\mathrm{LI}}$.

A Projected Spectrum

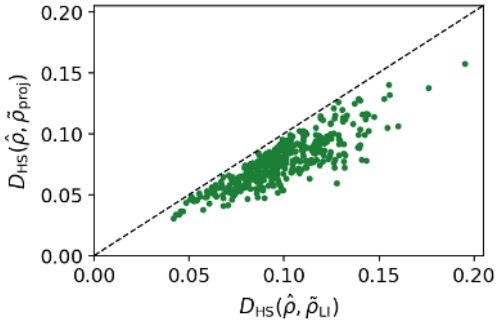


our simulation

Spectrum of the Bell-state estimate with the most negative eigenvalue at $M = 200$, before and after the projection, true spectrum in black.

The negative eigenvalues go to zero and the largest one is lowered.

The Projection Helps



our simulation

D_{HS} to the true state after against before the projection, 400 data sets.

Every point lies below the diagonal, as the convexity argument guarantees for D_{HS}. The trace distance also decreased, in all 400 data sets.

Maximum Likelihood

Maximum likelihood (MLE) searches for the physical state under which the counts are most probable,

$$\ell(\hat{\rho}) = \sum_{s,\mathbf{b}} n_s(\mathbf{b}) \ln \mathrm{Tr}\big[\hat{\rho}\,\hat{\pi}_{s,\mathbf{b}}\big], \qquad \tilde{\rho}_{\mathrm{MLE}} = \arg \max_{\hat{\rho}\geq 0,\ \mathrm{Tr}\hat{\rho}=1} \ell(\hat{\rho}).$$

A step from $\hat{\rho}$ towards a physical state $\hat{\sigma}$ changes it at the rate

$$\frac{d}{d\epsilon}\,\ell\big((1-\epsilon)\hat{\rho}+\epsilon\hat{\sigma}\big)\Big|_{\epsilon=0} = \sum_{s,\mathbf{b}} \frac{n_s(\mathbf{b})\,\big(\mathrm{Tr}[\hat{\sigma}\hat{\pi}_{s,\mathbf{b}}] - \mathrm{Tr}[\hat{\rho}\hat{\pi}_{s,\mathbf{b}}]\big)}{p_s(\mathbf{b}|\hat{\rho})} = 3^N M\big[\mathrm{Tr}(\hat{R}\hat{\sigma}) - 1\big],$$

$$\hat{R}(\hat{\rho}) = \frac{1}{3^N} \sum_{s,\mathbf{b}} \underbrace{\frac{f_s(\mathbf{b})}{p_s(\mathbf{b}|\hat{\rho})}}_{\text{data over model}} \hat{\pi}_{s,\mathbf{b}}, \qquad \mathrm{Tr}(\hat{R}\hat{\rho}) = 1.$$

At the maximum no direction increases ℓ, so $\lambda_{\max}(\hat{R}) = 1$ and $\hat{R}\tilde{\rho}_{\mathrm{MLE}} = \tilde{\rho}_{\mathrm{MLE}}$.

Z. Hradil, PRA **55**, R1561 (1997), doi:10.1103/PhysRevA.55.R1561

Two Algorithms for the Maximum

The fixed-point condition suggests the $R\rho R$ iteration, started at $\hat{I}/d$,

$$\hat{\rho}_{t+1} = \frac{\hat{R}(\hat{\rho}_t)\,\hat{\rho}_t\,\hat{R}(\hat{\rho}_t)}{\mathrm{Tr}\big[\hat{R}(\hat{\rho}_t)\,\hat{\rho}_t\,\hat{R}(\hat{\rho}_t)\big]},$$

where t counts the iterations. Every iterate is physical, and the diluted variant $(\hat{I}+\epsilon\hat{R})\hat{\rho}(\hat{I}+\epsilon\hat{R})$ never decreases ℓ for small enough ϵ.

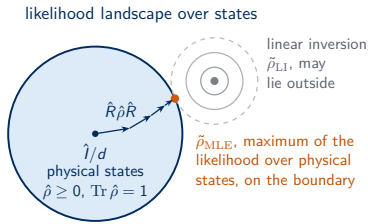
The second algorithm writes the state through a lower-triangular complex matrix C with real diagonal, a Cholesky factor,

$$\hat{\rho}(C) = \frac{CC^\dagger}{\mathrm{Tr}(CC^\dagger)},$$

which is physical for every C. Gradient descent on its d^2 real entries with Adam needs no constraint.

J. Řeháček *et al.*, PRA **75**, 042108 (2007), doi:10.1103/PhysRevA.75.042108 K. Banaszek *et al.*, PRA **61**, 010304 (1999), doi:10.1103/PhysRevA.61.010304 D. P. Kingma and J. Ba, arXiv:1412.6980

Likelihood over the Physical States



log-likelihood of the counts

$$\ell(\hat{\rho}) = \sum_{s,\mathbf{b}} n_s(\mathbf{b}) \ln \mathrm{Tr}\left[\hat{\rho}\,\hat{\pi}_{s,\mathbf{b}}\right]$$

$\hat{R}\hat{\rho}\hat{R}$ iteration

$$\hat{R} = \sum_{s,\mathbf{b}} \frac{f_s(\mathbf{b})}{\mathrm{Tr}[\hat{\rho}_t \hat{\pi}_{s,\mathbf{b}}]}\,\hat{\pi}_{s,\mathbf{b}}$$

$$\hat{\rho}_{t+1} = \frac{\hat{R}\hat{\rho}_t\hat{R}}{\mathrm{Tr}[\hat{R}\hat{\rho}_t\hat{R}]}$$

observed frequencies $f_s(\mathbf{b})$, constant factors cancel, every iterate is physical

Cholesky route $\hat{\rho} = \dfrac{CC^\dagger}{\mathrm{Tr}(CC^\dagger)}$

every lower-triangular C gives a physical state, so gradient descent on C needs no constraint

When the unconstrained optimum lies outside, the most likely physical state sits on the boundary and has a zero eigenvalue.

Error and the Number of Shots

Every frequency has a statistical error proportional to $M^{-1/2}$, and so has every linear-inversion coefficient. Distances that are linear in the error of the estimate therefore decrease as

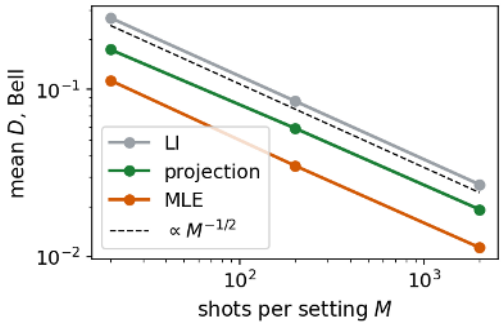
$$D \propto D_{\mathrm{HS}} \propto M^{-1/2}.$$

The infidelity of a physical estimate of a pure state $|\psi\rangle$ is the weight of the estimate outside $|\psi\rangle$,

$$1 - F = 1 - \langle\psi|\tilde{\rho}|\psi\rangle \approx \underbrace{\sin^2\delta}_{\text{large eigenvector rotated by } \delta \propto M^{-1/2}} + \underbrace{\textstyle\sum_{i\geq 2} \tilde{\lambda}_i}_{\text{small eigenvalues}} .$$

The first term falls as M^{-1}. For the MLE of the Bell state, an eigenstate of three measured settings, the second term falls as M^{-1} too, so $1 - F \propto M^{-1}$. For Haar-random pure states in the same settings we measure the slope -0.53. The projected estimate keeps eigenvalues of the size of the noise, $O(M^{-1/2})$, outside $|\psi\rangle$, and its infidelity falls only as $M^{-1/2}$. For full-rank states the M^{-1} law holds only asymptotically.

Trace Distance Against the Number of Shots

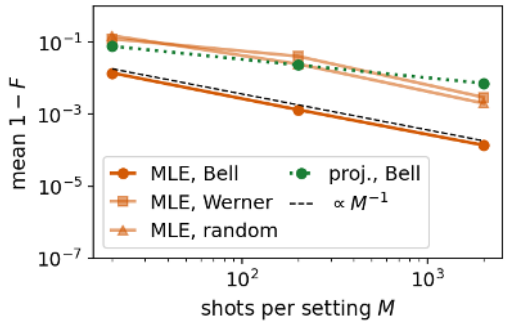


our simulation

Mean trace distance of the three Bell-state estimators over 400 data sets per point, with the line $M^{-1/2}$.

All three fall as $M^{-1/2}$, and MLE has the lowest error.

Infidelity Against the Number of Shots



our simulation

Mean infidelity of the MLE of the three states and of the projected Bell estimate, with the lines $M^{-1/2}$ and M^{-1}.

The Bell-state MLE falls with the slope -1.00, the projected estimate as $M^{-1/2}$. The Werner MLE has the slope -0.81 over this range of M.

The Cost Wall

The mean squared error of linear inversion follows from $\mathrm{Var}(\tilde{r}_P)$ and the number $\binom{N}{k}3^k$ of strings of weight k,

$$\mathbb{E}\, D_{\mathrm{HS}}^2 = \frac{1}{2^N} \sum_{P \neq I} \mathrm{Var}(\tilde{r}_P) \leq \frac{1}{2^N M} \sum_{k=1}^{N} \binom{N}{k} \frac{3^k}{3^{N-k}} = \frac{10^N - 1}{6^N M},$$

with equality for the maximally mixed state. With the total budget $M_{\mathrm{tot}} = 3^N M$,

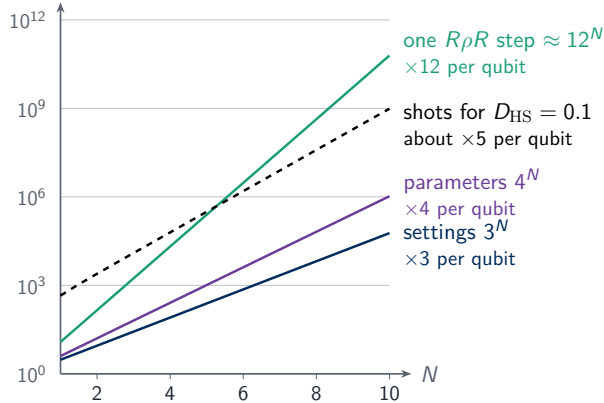
$$\mathbb{E}\, D_{\mathrm{HS}}^2 \approx \frac{5^N}{M_{\mathrm{tot}}} \qquad (N \gg 1),$$

so at a fixed total number of shots every added qubit multiplies the squared error by about five.

Settings grow as 3^N and parameters as 4^N. Building $\hat{R}$ costs of order $3^N 4^N = 12^N$ per $R\rho R$ step with the product structure, and the matrix products add 8^N. Larger systems need assumptions such as low rank or a smaller goal.

D. Gross *et al.*, PRL **105**, 150401 (2010), doi:10.1103/PhysRevLett.105.150401

Resources of Full Tomography



at a fixed total number of shots each setting receives only a fraction 3^{-N} of them

Ten qubits already need about 10^9 shots for $D_{\mathrm{HS}} = 0.1$, so full tomography stays limited to a few qubits.

3

A Network as a Denoising Filter



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Three Uses of Neural Networks in Tomography

- A network represents the **wave function itself**, for example a restricted Boltzmann machine trained to reproduce the measured bit strings. It describes structured states of many qubits, provided the state belongs to the family the network represents.
- A network maps the **measured frequencies directly** to a state. This removes the standard estimator and needs many training states.
- A network **corrects a standard estimate**. Linear inversion or maximum likelihood does the reconstruction, and the network removes the error that the finite number of shots leaves in it.

We develop the third use.

G. Torlai *et al.*, Nat. Phys. **14**, 447 (2018), doi:10.1038/s41567-018-0048-5 D. Koutný *et al.*, PRA **106**, 012409 (2022), doi:10.1103/PhysRevA.106.012409

Physical Estimates Are Biased

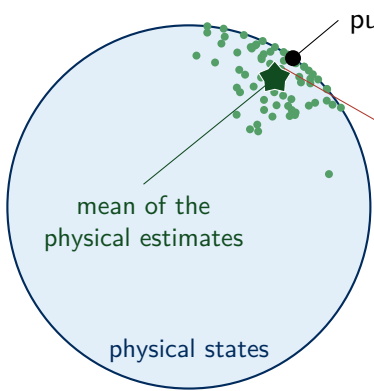
Linear inversion is unbiased, since it is linear in the frequencies, but it is unphysical. Projection and maximum likelihood return physical states, and their average over repetitions of the experiment misses the true state.

- A pure state is an **extreme point** of the convex set of density matrices, which means that it is no mixture of other states.
- A mean of physical estimates therefore equals a pure state only if every estimate equals it.
- Any spread of the estimates moves their mean towards mixed states, and the errors of many repetitions do not average out.

This systematic error is a **bias**. It is largest near the boundary of the state space and affects every estimator that always returns a physical state, maximum likelihood included.

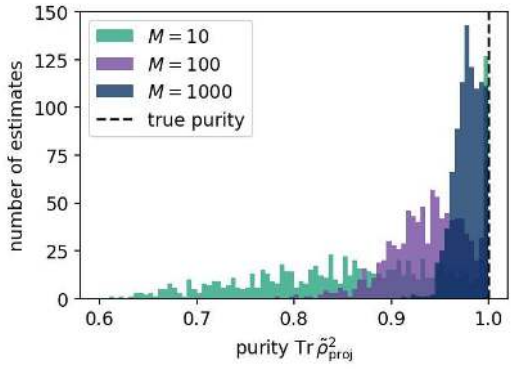
A. Macarone Palmieri *et al.*, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248

The Mean of Physical Estimates



projected estimates (dots) lie inside
or on the surface, so their mean
lies inside the ball

Purity of Projected Estimates



our simulation

Purity $\mathrm{Tr}\tilde{\rho}^2_{\mathrm{proj}}$ of the projected estimates of 1000 Haar-random pure states of two qubits, distributed uniformly over all pure states, for $M = 10$, 100 and 1000 shots per setting. The true purity is one.

Only a fraction 0.21, 0.11 and 0.08 of the estimates is pure. The mean purity, 0.88 at $M = 10$, approaches one with M but stays on one side of it.

The Denoising Protocol

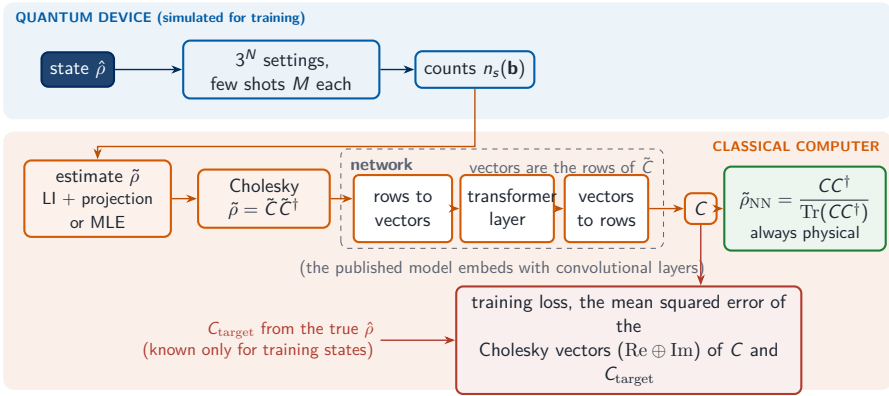
A network trained on pairs of a noisy and a clean photograph removes noise from new photographs. Here the noisy image is the standard estimate $\tilde{\rho}$ from few shots, and the clean image is the true state $\hat{\rho}$.

1. **Measure** the state in an informationally complete set of settings and record the frequencies.
2. **Pre-process** with a standard method, linear inversion with projection or maximum likelihood, which gives $\tilde{\rho}$.
3. **Post-process** with the network, which maps $\tilde{\rho}$ to a corrected physical state $\tilde{\rho}_{\mathrm{NN}}$.

Training data are simulated records of random states, Haar-random pure and Hilbert–Schmidt mixed, for which the true state is known. With a squared-error loss the network approximates the mean true state among training states with similar estimates, so it removes the bias conditioned on the estimate.

A. Macarone Palmieri *et al.*, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248

The Network Corrects an Estimate



The network acts on the Cholesky factor of the estimate, defined next.

4

Constructing the Loss



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Requirements on the Loss

The loss compares the output $\tilde{\rho}_{\mathrm{NN}}$ with the true state $\hat{\rho}$ for every training state at every step, and its gradient drives the training.

1. **Physical meaning.** A small loss must imply that the two states are hard to distinguish by any measurement.
2. **Low cost.** A state of dimension d has d^2 entries, and the loss should need little more than one pass over them.
3. **Smooth gradient.** The derivative with respect to the output must stay bounded, also when the output approaches the target.

We examine three distances against these requirements, the Bures distance, which follows from the geometry of the Hilbert space, the trace distance and the Hilbert–Schmidt distance.

M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (CUP, 2010), doi:10.1017/CBO9780511976667

The Distance Between Pure States

$|\psi\rangle$ and $e^{i\alpha}|\psi\rangle$ describe the same state, so a state is a **ray** in the Hilbert space. The distance between two states is the shortest distance between their representatives,

$$D^2(\psi,\phi) = \min_\alpha \big\| \,|\psi\rangle - e^{i\alpha}|\phi\rangle\big\|^2 = 2 - 2\underbrace{\max_\alpha \mathrm{Re}\big(e^{i\alpha}\langle\psi|\phi\rangle\big)}_{|\langle\psi|\phi\rangle|} = 2 - 2\,|\langle\psi|\phi\rangle|.$$

- It is zero only for the same state and largest, $D^2 = 2$, for orthogonal states.
- It depends only on the overlap $|\langle\psi|\phi\rangle|$, whose square is the fidelity F of pure states.
- The angle $\arccos|\langle\psi|\phi\rangle|$ between the rays is the Fubini–Study distance, for one qubit half the angle between the Bloch vectors.

I. Bengtsson and K. Życzkowski, *Geometry of Quantum States* (CUP, 2017), doi:10.1017/9781139207010

The Bures Distance Between Mixed States

A mixed state has **amplitudes**, matrices A with $\hat{\rho} = AA^\dagger$, for a pure state the matrix with $|\psi\rangle$ in one column. The freedom $A \to A\hat{U}$, with $\hat{U}$ unitary, replaces the phase. The Bures distance is again the shortest distance,

$$D_B^2(\hat{\rho},\hat{\sigma}) = \min_{\substack{\hat{\rho}=AA^\dagger \\ \hat{\sigma}=BB^\dagger}} \underbrace{\mathrm{Tr}\big[(A-B)(A-B)^\dagger\big]}_{\text{Euclidean distance of amplitudes}} = 2-2\sqrt{F(\hat{\rho},\hat{\sigma})}, \qquad \sqrt{F} = \mathrm{Tr}\sqrt{\sqrt{\hat{\rho}}\,\hat{\sigma}\,\sqrt{\hat{\rho}}}.$$

- **Meaning.** F generalises the overlap, and for pure states D_B is the distance between rays. No physical operation increases D_B.
- **Cost.** Two matrix square roots, each a diagonalisation with about d^3 operations.
- **Gradient.** The derivative of a matrix square root diverges for eigenvalues near zero, as in nearly pure states.

Remark. The published paper writes D_B for $2-2\sqrt{F}$, which is the squared Bures distance.

A. Uhlmann, Rep. Math. Phys. **9**, 273 (1976), doi:10.1016/0034-4877(76)90060-4

The Trace Distance

$$D_{\mathrm{tr}}(\hat{\rho},\hat{\sigma}) = \tfrac{1}{2}\sum_{i=1}^{d} \underbrace{|\lambda_i|}_{\text{eigenvalues of } \hat{\rho}-\hat{\sigma}} = \underbrace{\max_{\hat{E}} \left|\mathrm{Tr}[\hat{E}\hat{\rho}] - \mathrm{Tr}[\hat{E}\hat{\sigma}]\right|}_{\text{largest difference of an outcome probability}} .$$

- **Meaning.** The maximum runs over all outcomes, operators $0 \le \hat{E} \le \hat{I}$. Given either state with equal probability, the best measurement identifies it with probability $\frac{1}{2}(1 + D_{\mathrm{tr}})$.
- **Properties.** No physical operation increases it, and $1 - \sqrt{F} \le D_{\mathrm{tr}} \le \sqrt{1-F}$.
- **Cost.** A diagonalisation, about d^3 operations per training state and step.
- **Gradient.** $|\lambda_i|$ has a kink at $\lambda_i = 0$, where a good reconstruction ends.

C. W. Helstrom, J. Stat. Phys. **1**, 231 (1969), doi:10.1007/BF01007479 C. A. Fuchs and J. van de Graaf, IEEE Trans. Inf. Theory **45**, 1216 (1999), doi:10.1109/18.761271

The Hilbert–Schmidt Distance

$$D_{\mathrm{HS}}^2(\hat{\rho},\hat{\sigma}) = \mathrm{Tr}\left[(\hat{\rho}-\hat{\sigma})^2\right] = \sum_{a,b=1}^{d} \left|\rho_{ab}-\sigma_{ab}\right|^2 = \underbrace{\frac{1}{2^N}\sum_{\hat{P}} \left(\langle\hat{P}\rangle_\rho - \langle\hat{P}\rangle_\sigma\right)^2}_{N \text{ qubits, all } 4^N \text{ Pauli strings } \hat{P}}.$$

- **Meaning.** The Euclidean distance of the matrices. It sums the squared differences of all Pauli expectation values, so it measures the difference for an average observable, whereas the trace distance measures the best one.
- **Properties.** A physical operation can increase it, and $D_{\mathrm{HS}} \leq 2D_{\mathrm{tr}} \leq \sqrt{d}\, D_{\mathrm{HS}}$, a factor that grows with the dimension.
- **Cost and gradient.** One sum over the d^2 entries, smooth everywhere. It is the squared error of neural-network libraries.

M. Ozawa, Phys. Lett. A **268**, 158 (2000), doi:10.1016/S0375-9601(00)00171-7

Three Distances Compared

	Bures	Trace	Hilbert–Schmidt
Meaning	distance of rays, overlap of the states	difference for the best measurement	difference for an average observable
Never increased by physical operations	yes	yes	no
Cost per state	two diagonalisations, $\sim d^3$	diagonalisation, $\sim d^3$	one sum, $\sim d^2$
Gradient	diverges near pure states	kinks	smooth

A Loss from One Fixed Pair of Amplitudes

No distance meets all three requirements when applied to the states themselves.

The Bures distance is a minimum of Euclidean distances over all amplitudes. The published loss skips the minimisation and uses one fixed pair of amplitudes of unit trace, the **Cholesky factors** of the two states. The distance of a fixed pair is at least the minimum,

$$\underbrace{\left\|\mathbf{c}_1 - \mathbf{c}_2\right\|^2}_{\substack{\text{squared error of Cholesky vectors,} \\ \text{cheap and smooth}}} \quad \geq \quad \underbrace{D_B^2(\hat{\rho}, \hat{\sigma})}_{\substack{\text{squared Bures distance,} \\ \text{physical meaning}}} \quad .$$

Minimising the left-hand side forces the fidelity up, and through $D_{\mathrm{tr}} \leq \sqrt{1 - F}$ it also controls the trace distance.

The Cholesky factor, its vector $\mathbf{c}$ and the proof of the inequality follow.

The Cholesky Representation

A network outputs unconstrained numbers, and a density matrix must be non-negative. Every positive definite matrix factorises uniquely as

$$\hat{\rho} = CC^\dagger,$$

with a lower-triangular **Cholesky factor** C with real positive diagonal (for pure states we factorise $(\hat{\rho} + \epsilon\hat{I})/(1 + d\epsilon)$, $\epsilon = 10^{-6}$). Conversely every complex C defines a physical state,

$$\tilde{\rho}_{\mathrm{NN}} = \frac{CC^\dagger}{\operatorname{Tr} CC^\dagger}.$$

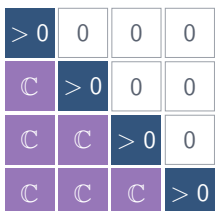
The free entries form the Cholesky vector of $d(d+1)/2 + d(d-1)/2 = d^2$ real numbers,

$$\mathbf{c} = \underbrace{(\operatorname{Re} C_{ij})_{i \geq j}}_{\text{lower triangle and diagonal}} \oplus \underbrace{(\operatorname{Im} C_{ij})_{i > j}}_{\text{strict lower triangle}},$$

16 numbers for two qubits and 256 for four.

Factor, Vector and State

Cholesky factor C, $d = 4$



real positive diagonal
complex below the diagonal
zeros above

Cholesky vector $\mathbf{c}$, $d^2 = 16$ reals



$\operatorname{Re} C_{ij},\ i \geq j$ — 10 numbers
$\operatorname{Im} C_{ij},\ i > j$ — 6 numbers

$$\tilde{\rho} = \frac{CC^\dagger}{\operatorname{Tr} CC^\dagger}$$

physical for every C

The Shortest Distance Gives the Fidelity

For states of unit trace $\mathrm{Tr}[(A-B)(A-B)^\dagger] = 2 - 2\,\mathrm{Re}\,\mathrm{Tr}AB^\dagger$, so the shortest distance needs the largest $\mathrm{Re}\,\mathrm{Tr}AB^\dagger$. Every amplitude of $\hat\rho$ is $A = \sqrt{\hat\rho}\,\hat V$ with a unitary $\hat V$ (polar decomposition). For amplitudes A of $\hat\rho$ and $B = \sqrt{\hat\sigma}\,\hat W$ of $\hat\sigma$,

$$\mathrm{Tr}\,AB^\dagger = \mathrm{Tr}(\sqrt{\hat\rho}\,\hat V\hat W^\dagger\sqrt{\hat\sigma}) = \mathrm{Tr}(\underbrace{\sqrt{\hat\sigma}\sqrt{\hat\rho}}_{Q}\ \underbrace{\hat V\hat W^\dagger}_{\text{unitary }\hat U}).$$

For any matrix Q and unitary $\hat U$, $|\mathrm{Tr}(Q\hat U)|$ is at most the sum of the singular values of Q, and a suitable $\hat U$ reaches it. Here

$$\mathrm{Re}\,\mathrm{Tr}\,AB^\dagger \;\le\; \mathrm{Tr}\sqrt{Q^\dagger Q} = \mathrm{Tr}\sqrt{\sqrt{\hat\rho}\,\hat\sigma\sqrt{\hat\rho}} = \sqrt{F(\hat\rho,\hat\sigma)},$$

$$\sqrt{F(\hat\rho,\hat\sigma)} = \max_{\hat\rho=AA^\dagger,\ \hat\sigma=BB^\dagger} \tfrac{1}{2}\mathrm{Tr}\left[AB^\dagger + BA^\dagger\right].$$

For pure states the maximum is $|\langle\psi|\phi\rangle|$. The notes prove both statements used here.

A. Uhlmann, Rep. Math. Phys. **9**, 273 (1976), doi:10.1016/0034-4877(76)90060-4

The Cholesky Factors Give a Bound

The Cholesky factors C_1 of $\hat{\rho}$ and C_2 of $\hat{\sigma}$ are one particular pair of amplitudes, which need not reach the maximum,

$$\sqrt{F(\hat{\rho},\hat{\sigma})} \;\geq\; \tfrac{1}{2}\mathrm{Tr}\big[C_1 C_2^\dagger + C_2 C_1^\dagger\big].$$

Both states have unit trace, so $2 = \mathrm{Tr}\hat{\rho} + \mathrm{Tr}\hat{\sigma} = \mathrm{Tr}\, C_1 C_1^\dagger + \mathrm{Tr}\, C_2 C_2^\dagger$, and

$$D_B^2(\hat{\rho},\hat{\sigma}) = \underbrace{2}_{\mathrm{Tr}C_1C_1^\dagger + \mathrm{Tr}C_2C_2^\dagger} - 2\sqrt{F} \leq \mathrm{Tr}\big[C_1C_1^\dagger + C_2C_2^\dagger - C_1C_2^\dagger - C_2C_1^\dagger\big]$$

$$= \mathrm{Tr}\big[(C_1 - C_2)(C_1 - C_2)^\dagger\big] \;=\; D_{\mathrm{HS}}^2(C_1, C_2).$$

The squared Bures distance of two states is bounded by the **Hilbert–Schmidt distance of their Cholesky factors**, the Euclidean distance of the two matrices.

The Distance of the Factors Is a Squared Error

With $X = C_1 - C_2$ and the Cholesky vectors $\mathbf{c}_1$, $\mathbf{c}_2$ of real and imaginary parts,

$$D^2_{\mathrm{HS}}(C_1, C_2) = \operatorname{Tr} XX^\dagger = \sum_{a,b} |X_{ab}|^2 = \sum_{a,b} \left[(\operatorname{Re} X_{ab})^2 + (\operatorname{Im} X_{ab})^2\right] = \|\mathbf{c}_1 - \mathbf{c}_2\|^2.$$

The loss is this squared error between the network output and the Cholesky vector of the true state, averaged over a batch of N_b training states,

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N_b} \sum_{m=1}^{N_b} \left\| \mathbf{c}_{\boldsymbol{\theta}}(\tilde{C}_m) - \mathbf{c}_{\mathrm{target},m} \right\|^2 \;\geq\; \frac{1}{N_b} \sum_{m=1}^{N_b} D_B^2(\tilde{\rho}_{\mathrm{NN},m}, \hat{\rho}_m).$$

A small loss guarantees a high fidelity, $F \geq \left(1 - \frac{1}{2}\|\mathbf{c}_1 - \mathbf{c}_2\|^2\right)^2$ for each state.

Remarks. The bound needs factors of unit trace, so our classroom loss normalises the output to $\operatorname{Tr} CC^\dagger = 1$. The published loss adds the regulariser $\operatorname{Tr}[C_{\boldsymbol{\theta}} C_{\boldsymbol{\theta}}^\dagger]$ and normalises the state after the network.

A. Macarone Palmieri *et al.*, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248

5
Architecture



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The Published Network

The network maps the Cholesky vector $\tilde{\mathbf{c}}$ of the estimate to a corrected one,

$$\mathbf{c}_{\boldsymbol{\theta}} = \underbrace{\tanh\left(h_{\mathrm{l}}\right)}_{\substack{\text{output layer,}\\ \text{back to } d^2 \text{ numbers}}} \circ \underbrace{h_{\mathrm{tr}}}_{\substack{\text{transformer layer}\\ \text{with attention}}} \circ \underbrace{\gamma\left(h_{\mathrm{cnn}}\right)}_{\substack{\text{convolutional layers}\\ \text{with activation } \gamma \text{ (GELU)}}} \left[\tilde{\mathbf{c}}\right].$$

- The **convolutional layers** extract local features and return K feature vectors.
- The **transformer layer** mixes these K vectors by attention, which relates every vector to every other.
- The **output layer** combines the result into the d^2 numbers of a Cholesky vector.

Remark. In the published network the vectors that enter attention are the K feature vectors of the convolutions. Our classroom network (Lab 076) has no convolutions and uses the d rows of $\tilde{C}$ instead.

A. Macarone Palmieri *et al.*, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248

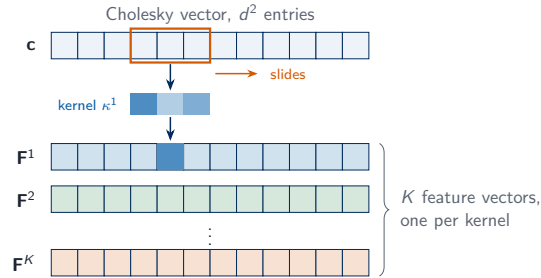
D. Hendrycks and K. Gimpel, arXiv:1606.08415

A Convolutional Layer on the Cholesky Vector

A kernel is a vector of w trained weights that slides along the input. At position p it returns

$$F_p^k = \gamma\Big(\sum_{q=1}^{w} \underbrace{\kappa_q^k}_{\text{trained}} \; c_{p+q} + b^k \Big).$$

- K kernels give K feature vectors $\mathbf{F}^1, \ldots, \mathbf{F}^K$.
- Each output sees a **window** of w neighbouring entries.
- The weights are the **same** at every position and for every input.



6

The Attention Mechanism

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Couplings That Depend on the Data

The input is a set of K vectors $\mathbf{x}_1, \ldots, \mathbf{x}_K$ that describe parts of one object, here features of the Cholesky factor of a noisy estimate. A linear layer mixes them with **fixed couplings**,

$$\mathbf{y}_i = \sum_{j=1}^{K} \underbrace{J_{ij}}_{\substack{\text{trained, then the same} \\ \text{for every input}}} \mathbf{x}_j .$$

Attention mixes them with **couplings computed from the vectors themselves**,

$$\mathbf{y}_i = \sum_{j=1}^{K} \underbrace{a_{ij}(\mathbf{x}_1, \ldots, \mathbf{x}_K)}_{\substack{\text{recomputed for} \\ \text{every input}}} \underbrace{W_V \mathbf{x}_j}_{\text{content of part } j} .$$

The weights a_{ij} follow from three operations, which compare the vectors, normalise the result and mix.

D. Bahdanau, K. Cho and Y. Bengio, arXiv:1409.0473 A. Vaswani *et al.*, NeurIPS **30** (2017), arXiv:1706.03762

A Trained Rule for Comparing Two Vectors

The dot product $\mathbf{x}_i \cdot \mathbf{x}_j$ measures how well two vectors align, but the rule is fixed. Attention inserts two trained matrices W_{L} and W_{R} with d_h rows and compares the images of the two vectors,

$$s_{ij} = (W_{\mathrm{L}}\mathbf{x}_i) \cdot (W_{\mathrm{R}}\mathbf{x}_j) = \mathbf{x}_i^T \underbrace{W_{\mathrm{L}}^T W_{\mathrm{R}}}_{g} \mathbf{x}_j = \mathbf{x}_i^T \, g \, \mathbf{x}_j.$$

- The matrix g is a **trained bilinear form**. With $g = \hat{I}$ the score is the ordinary dot product.
- A large score s_{ij} means that part j is relevant for updating part i.
- g has rank at most d_h, so the comparison uses only d_h directions of the vector space.
- After training g is the same for every input. The scores s_{ij} change with the input.

Scores Become Boltzmann Weights

For each i the scores become positive weights that sum to one, a normalisation called softmax,

$$a_{ij} = \frac{\exp\left(s_{ij}/\sqrt{d_h}\right)}{\underbrace{\sum_{j'=1}^{K} \exp\left(s_{ij'}/\sqrt{d_h}\right)}_{Z_i}}, \qquad \sum_{j=1}^{K} a_{ij} = 1.$$

They form a canonical distribution over the parts j, with energies $E_{ij} = -s_{ij}$, temperature $\sqrt{d_h}$ and partition function Z_i.

- Scores that differ by much more than $\sqrt{d_h}$ put all weight on the best-matching part, as at low temperature.
- Nearly equal scores give a uniform average over all parts, as at high temperature.
- Through Z_i the weight of part j depends on all other parts of the same input.

A Weighted Average of the Content

A third trained matrix W_{V} selects what each part passes on, and the output is the average with these weights,

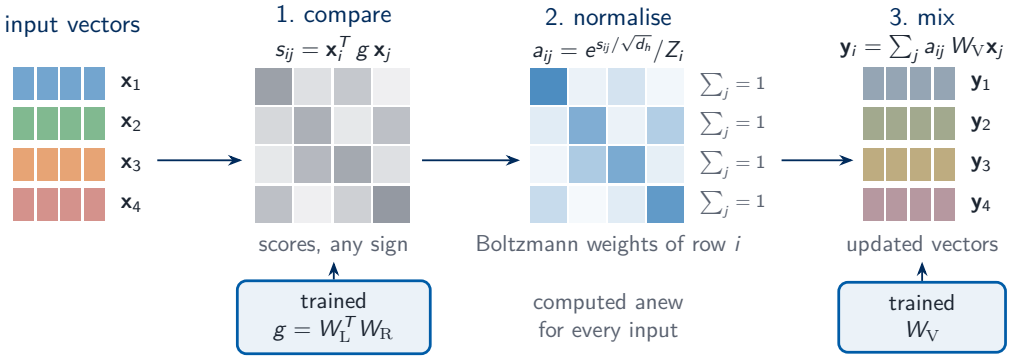
$$\mathbf{y}_i = \sum_{j=1}^{K} a_{ij}\, W_{\mathrm{V}}\, \mathbf{x}_j,$$

a thermal average of the transformed vectors.

Trained, the same for every input	**Computed anew for every input**
W_{L}, W_{R} (the rule of comparison g)	scores s_{ij}
W_{V} (the content that is mixed)	weights a_{ij}
	output vectors $\mathbf{y}_i$

The three matrices are trained like all other weights, by gradient descent on the loss of the whole network.

Compare, Normalise, Mix



The Bilinear Form g Matches Features

The singular value decomposition $g = \sum_a \sigma_a \, \mathbf{p}_a \mathbf{q}_a^T$ writes the score as a sum over d_h pairs of directions,

$$s_{ij} = \mathbf{x}_i^T \, g \, \mathbf{x}_j = \sum_{a=1}^{d_h} \sigma_a \underbrace{(\mathbf{p}_a \cdot \mathbf{x}_i)}_{\substack{\text{feature } a \text{ of the} \\ \text{receiving part } i}} \underbrace{(\mathbf{q}_a \cdot \mathbf{x}_j)}_{\substack{\text{feature } a \text{ of the} \\ \text{source part } j}} .$$

- g decides **which feature of one part is matched with which feature of another**.
- $g = \hat{I}$ matches every direction only with itself, which is plain similarity. $\mathbf{p}_a \neq \mathbf{q}_a$ matches different features, so a part can draw on parts unlike itself.
- Training chooses the directions that lower the loss. Here they encode which features make one part of the Cholesky factor useful for correcting another.

A Trained Metric for Equal Matrices

For $W_{\mathrm{L}} = W_{\mathrm{R}} = W$ the form $g = W^T W$ is symmetric and positive semidefinite. With $\mathbf{u}_i = W\mathbf{x}_i$,

$$s_{ij} = \mathbf{u}_i \cdot \mathbf{u}_j = \tfrac{1}{2}|\mathbf{u}_i|^2 + \tfrac{1}{2}|\mathbf{u}_j|^2 - \tfrac{1}{2}\underbrace{|\mathbf{u}_i - \mathbf{u}_j|^2}_{d_g^2(\mathbf{x}_i,\mathbf{x}_j)}, \qquad d_g^2(\mathbf{x},\mathbf{x}') = (\mathbf{x}-\mathbf{x}')^T g\,(\mathbf{x}-\mathbf{x}').$$

The term $|\mathbf{u}_i|^2$ cancels in the normalisation over j, which leaves

$$a_{ij} \propto \exp\Big[\frac{-d_g^2(\mathbf{x}_i,\mathbf{x}_j) + |\mathbf{u}_j|^2}{2\sqrt{d_h}}\Big],$$

a Gaussian weight in a **trained distance** d_g, times a factor that depends only on j. Close parts mix strongly.

In general $W_{\mathrm{L}} \neq W_{\mathrm{R}}$, and g is neither symmetric nor positive, so part i may use part j strongly while j ignores i. Attention trains a **rule of comparison**, which is a metric only for equal matrices.

Training Data Fix the Rule, the Input Fixes the Weights

- The **training data** fix the matrices g and W_{V}. They encode which features of two parts make them relevant to each other, here across many pairs of a noisy estimate and its true state.
- The **present input** fixes the weights a_{ij}. The same trained g gives different couplings for different estimates, and the weight of one part depends on all others through Z_i.
- With **several layers** the second layer compares vectors $\mathbf{y}_i$ that already contain the other parts. The effective comparison of two original parts then depends on the whole input.

Heads and the Transformer Layer

Several **heads** apply independent triples $W_{\mathrm{L}}, W_{\mathrm{R}}, W_{\mathrm{V}}$ in parallel, each with its own rule of comparison, and a linear map combines their outputs.

A **transformer layer** adds the attention output to the input, $\mathbf{x}_i + \mathbf{y}_i$, and then applies a small two-layer network to every vector separately.

A. Vaswani *et al.*, NeurIPS **30** (2017), arXiv:1706.03762

Convolution and Attention Compared

	Convolution	Attention
Range of one layer	a window of w neighbouring entries	every vector with every other
Couplings	kernel weights, fixed after training and the same for every input	weights a_{ij}, recomputed from every input
What is trained	the couplings themselves	the rule g that produces the couplings
As a noise filter	average over a fixed neighbourhood	average over parts that are similar under the trained comparison

Advantages of Attention over Convolutions

- **The relevant correlations are non-local.** An entry of the state couples two rows of the factor, $\rho_{ij} = \sum_l C_{il} C^*_{jl}$. In the flattened Cholesky vector these entries lie far apart, outside one window of a kernel. Entanglement resides in such coherences. The inputs of attention are the rows (classroom network) or feature vectors that extend over all positions (published network), so one layer relates all parts.

- **The best correction depends on the estimate.** A convolution applies the same average to every input. Attention averages over the parts that the data select, which is the principle of non-local means in image denoising. In the metric case the weights a_{ij} are Gaussian weights in the trained distance.

These two points are an interpretation. The evidence is the published comparison with networks of convolutions alone at an equal number of parameters.

A. Buades, B. Coll and J.-M. Morel, CVPR **2**, 60 (2005), doi:10.1109/CVPR.2005.38

7

Published Results



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Training on Synthetic Haar States

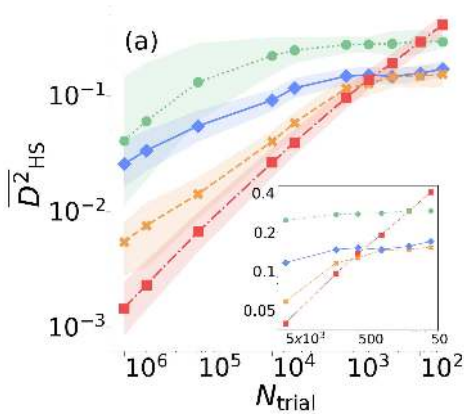
The network is trained on simulated data only.

1. **Draw** random target states $\hat{\rho}_m$. Haar-random pure states are distributed uniformly over all pure states.
2. **Simulate** the measurement of each state with a finite number of copies, which gives frequencies with shot noise.
3. **Reconstruct** each state by linear inversion with the negative eigenvalues removed, which gives the noisy Cholesky factor $\tilde{C}_m$.
4. **Train** on the pairs of a noisy factor $\tilde{C}_m$ and the true factor of $\hat{\rho}_m$ with the squared-error loss.

The published networks use between 2×10^3 and 10^4 such pairs. The first test uses new random states of the same kind. The second uses states of a different kind, which the network never saw.

A. Macarone Palmieri *et al.*, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248

Published Result: Fewer Shots for the Same Error



published result, panel (a), not our simulation

Random mixed states of dimension $d = 9$. Mean squared distance to the true state against the number of copies N_{trial}, which decreases to the right. The inset enlarges few copies.

green circles	linear inversion
red squares	maximum likelihood
blue diamonds	network on linear inversion
orange crosses	network on maximum likelihood

The network improves linear inversion at every number of copies. It improves maximum likelihood below about 10^3 copies, and with many copies maximum likelihood alone is best.

A. Macarone Palmieri *et al.*, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248

One-Axis Twisting as a Test Outside the Training Set

States made in experiments are not random, since an interaction produces them. The one-axis twisting protocol starts with all spins along x, $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and lets them interact through the collective spin $\hat{J}_z = \frac{1}{2}\sum_q \hat{\sigma}_{z,q}$,

$$|\Psi(t)\rangle = \underbrace{e^{-it\hat{J}_z^2}}_{\substack{\text{each spin precesses at a rate set} \\ \text{by the magnetisation of all others}}} |+\rangle^{\otimes N}.$$

- At short times the state is spin squeezed.
- At $t = \pi/2$ it is a cat state, all spins along $+x$ superposed with all along $-x$.

The network is trained on Haar-random states only and is then applied to estimates of these states, an inference **outside the training distribution**.

M. Kitagawa and M. Ueda, PRA **47**, 5138 (1993), doi:10.1103/PhysRevA.47.5138

The Quantum Fisher Information

To judge the reconstruction of an entangled state we need a quantity that detects its entanglement. A collective rotation by an angle θ about an axis $\mathbf{n}$ encodes θ in the state,

$$\hat{\rho}(\theta) = e^{-i\theta \hat{J}_{\mathbf{n}}}\, \hat{\rho}\, e^{i\theta \hat{J}_{\mathbf{n}}}.$$

The **quantum Fisher information** F_Q measures how strongly the state responds to a small change of θ,

$$\underbrace{D_B^2(\hat{\rho}(\theta), \hat{\rho}(\theta + d\theta))}_{\text{squared Bures distance of neighbouring states}} = \tfrac{1}{4} F_Q\, d\theta^2, \qquad \underbrace{\Delta\theta \geq \frac{1}{\sqrt{\mu F_Q}}}_{\text{precision with } \mu \text{ copies}} .$$

For a pure state $F_Q = 4\,\mathrm{Var}(\hat{J}_{\mathbf{n}})$. A sensitive state changes much under a small rotation and measures θ precisely.

S. L. Braunstein and C. M. Caves, PRL **72**, 3439 (1994), doi:10.1103/PhysRevLett.72.3439

Fisher Information Certifies Entanglement Depth

The sensitivity of a state to a collective rotation is limited by its entanglement.

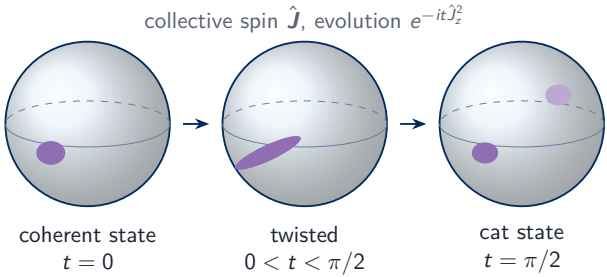
- A product state of N spins has $F_Q \leq N$.
- A state in which at most m spins are entangled with each other has $F_Q \leq mN$.
- The cat state, with all N spins entangled, reaches the maximum $F_Q = N^2$.

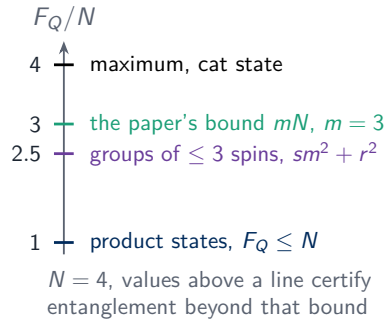
A measured value $F_Q/N > m$ therefore certifies an **entanglement depth** of at least $m+1$, which means that at least $m+1$ spins are entangled with each other. For four qubits $F_Q/N > 3$ certifies the entanglement of all four.

The Fisher information of a reconstructed state thus tells whether the reconstruction kept the entanglement of the true state.

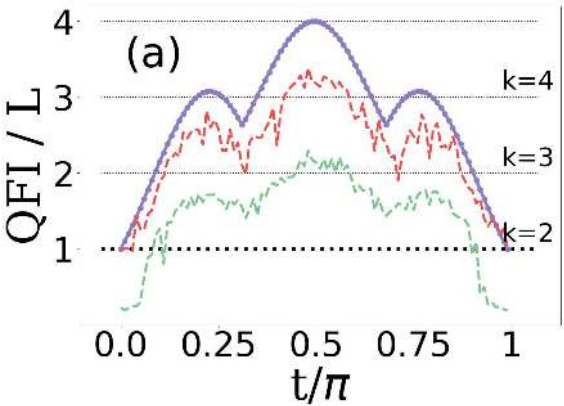
P. Hyllus *et al.*, PRA **85**, 022321 (2012), doi:10.1103/PhysRevA.85.022321 G. Tóth, PRA **85**, 022322 (2012), doi:10.1103/PhysRevA.85.022322

Twisting and the Fisher-Information Ladder





Published Result: Linear Inversion Plus Network



published result, panel (a), not our simulation

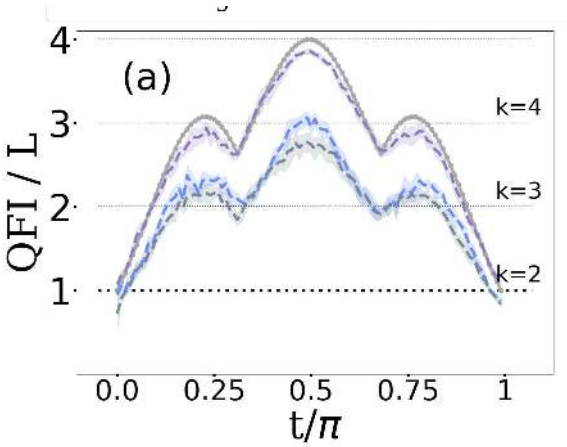
F_Q/N (written QFI/L) against the twisting time for noisy one-axis-twisted states of four qubits.

Violet, noiseless target states. Green, linear inversion. Red, network on linear inversion. Above the dotted line at 3, labelled $k = 4$ in the figure, all four spins are entangled.

Linear inversion stays below the line. With the network the reconstruction crosses it, as the target states do. A learned estimate has no error bound, so this indicates four-spin entanglement without certifying it.

A. Macarone Palmieri *et al.*, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248

Published Comparison of Convolutions and Attention



F_Q/N of the same one-axis-twisted states for post-processing networks with an equal number of parameters.

grey, solid	noiseless target states
violet, dashed	convolutions and attention
blue, dashed	two convolutional layers
dark, dashed	linear inversion alone

Convolutions alone stay close to linear inversion, and the network with attention follows the target states. Four convolutional layers, panel (b) of the source, give the same result.

published result, panel (a), not our simulation

A. Macarone Palmieri *et al.*, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248

8

Classroom Version



Two Classroom Networks

Both networks return the input factor plus a learned correction, $C_\theta = \tilde{C} + \Delta_\theta(\tilde{C})$, and the last layer starts at zero, so an untrained network returns the projected estimate.

The **multilayer perceptron** (MLP) treats the 16 numbers of $\mathbf{c}$ as one vector, with two hidden layers of 80 units and the smooth rectifier GELU. The **attention network** has no convolutions and uses the structure of the factor. An entry of the density matrix couples two rows $\mathbf{r}_i$ of C,

$$\rho_{ij} = \sum_l C_{il} C^*_{jl} = \mathbf{r}_i \cdot \mathbf{r}^*_j ,$$

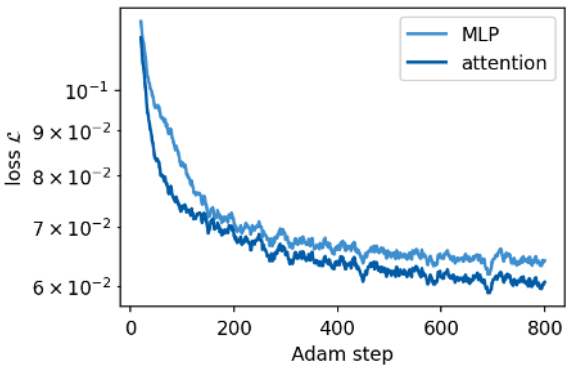
so the correction of one row depends on all others. Each row becomes one of the vectors that enter attention, of width $d_{\text{model}} = 32$,

$$\mathbf{x}_i = W_E \left(\operatorname{Re} \mathbf{r}_i, \operatorname{Im} \mathbf{r}_i \right) + \underbrace{\mathbf{e}_i}_{\text{position of the row}} ,$$

with a learned matrix W_E. Both networks have about 9000 parameters.

D. Hendrycks and K. Gimpel, arXiv:1606.08415

Training Two Qubits

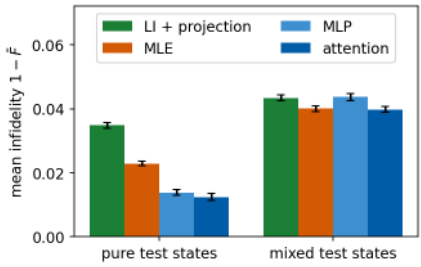


our simulation

Loss on mini-batches at $M = 100$ shots per setting, averaged over 20 steps.

5000 training states, half Haar and half Hilbert–Schmidt, 800 Adam steps with batches of 250. The loss starts at the error of the projected estimate and halves. The noise sets a floor, since the same estimate arises from many true states.

Fidelities at $M = 100$

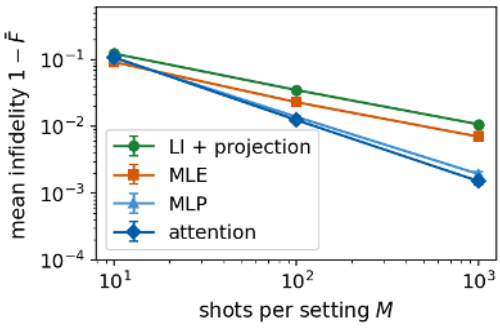


our simulation

Mean infidelity $1 - \bar{F}$ (bar for the mean over test states) over 500 pure and 500 mixed test states for the projected estimate, maximum likelihood (MLE) and the two networks, with standard errors.

For pure states both networks lower the mean infidelity well below that of MLE, and they agree with each other within the error bars. For mixed states, which lie far from the boundary and carry little bias, all four estimators agree within 0.005.

Pure Test States Across Shot Numbers

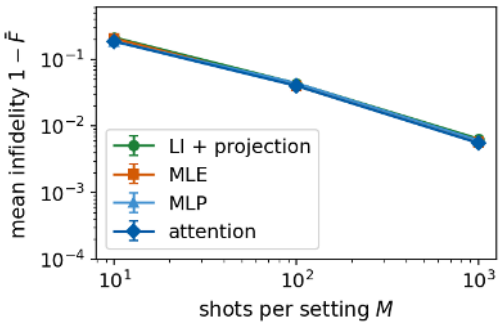


our simulation

Mean infidelity against M for pure test states. A separate network is trained for every M.

The networks beat MLE at $M = 100$ and 1000, whereas at $M = 10$ MLE is best.

Mixed Test States Across Shot Numbers



our simulation

Mean infidelity against M for mixed test states.

The networks gain only at $M = 10$. At larger M all four estimators agree within 0.005.

A Tighter Threshold for Four Qubits

We take the largest Fisher information over the rotation axes $\mathbf{n}$. Product states have $F_Q \le N$. A state built from groups of at most m entangled spins obeys

$$F_Q \le \underbrace{sm^2 + r^2}_{N = sm + r,\ 0 \le r < m} \le mN.$$

A measured F_Q above the bound for m therefore certifies a group of at least $m+1$ entangled spins.

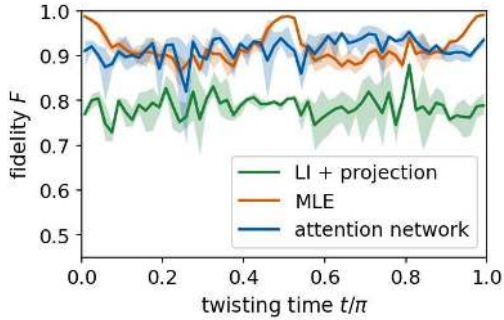
For $N = 4$ and $m = 3$ the first form gives $s = r = 1$ and $F_Q \le 10$, so

$$\frac{F_Q}{N} > 2.5 \quad \text{certifies entanglement of all four spins.}$$

The looser form mN gives the threshold $F_Q/N > 3$, which the published work uses. Both thresholds are sufficient conditions, and the cat state reaches the maximum $F_Q/N = 4$.

P. Hyllus *et al.*, PRA **85**, 022321 (2012), doi:10.1103/PhysRevA.85.022321 G. Tóth, PRA **85**, 022322 (2012), doi:10.1103/PhysRevA.85.022322

Four Qubits

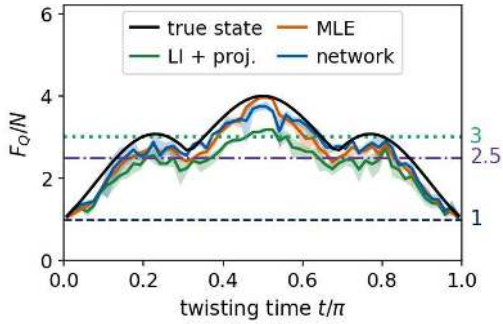


our simulation

Attention network, 47072 parameters, trained on Haar-random states only, $M = 10$ in each of 81 settings. Fidelity of the one-axis-twisted states, mean and range of three records. Orange, MLE.

The network raises the mean fidelity from 78.6 % to 91.5 %, although it never saw a twisted state. MLE reaches 91.8 %, the same within the error.

Fisher Information of the Reconstructions



our simulation

Largest F_Q/N of the true states, the projected estimates, maximum likelihood and the network outputs against the twisting time.

The projected estimates lose much of the Fisher information. The network and maximum likelihood recover a large part of it.

Entanglement of the Reconstructed States

For every record of the one-axis-twisted states we compute the largest F_Q/N of the true state, of the network output and of the projected estimate. The table gives the fractions of records above the two thresholds.

	true states	network	maximum likelihood	projected
$F_Q/N > 2.5$	0.73	0.66	0.59	0.37
$F_Q/N > 3$	0.43	0.27	0.22	0.09

The projected estimate loses much of the Fisher information, since its noise mixes the state, and reaches at most 3.22. The network recovers much of it, but it overestimates the true value in a fraction 0.16 of the records and has no error bound, so its values certify no entanglement by themselves.

9

Hamiltonian Learning

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Learning a Hamiltonian from Measured Bit Strings

Lecture 12 learned a Hamiltonian with a quantum circuit that had access to the evolved states. Here the device returns only bit strings, and the model runs on a classical computer.

	Lecture 12, QGRNN	Vanilla fit	Neural differential equation
Data	evolved states kept in the device	measured bit strings	measured bit strings
Model	trainable Trotter circuit	simulated assumed Hamiltonian	the same plus a network in a warm-up
Comparison	SWAP test of two states	likelihood of the bit strings	likelihood of the bit strings

The vanilla fit adjusts the coefficients by gradient descent, with gradients from automatic differentiation through the simulated evolution. We treat it first, as the plain fit.

G. Verdon *et al.*, arXiv:1909.12264 T. Heightman, E. Jiang and A. Acín, Quantum Sci. Technol. **10**, 045072 (2025), doi:10.1088/2058-9565/ae0d79

A Hamiltonian Known in Form

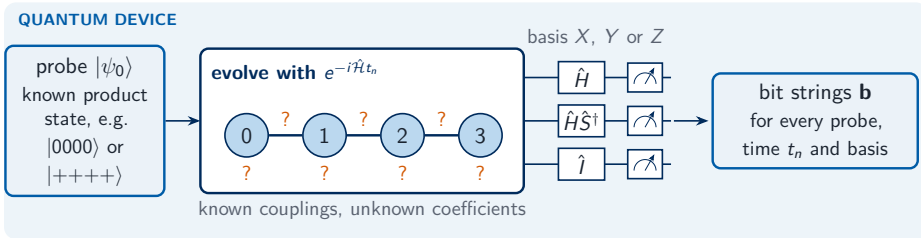
In a quantum simulator drifting calibrations set the strengths of the designed couplings and fields, so the Hamiltonian is known in form and unknown in its numbers. Our device is an anisotropic Heisenberg (XYZ) chain of $N = 4$ qubits in two uniform fields,

$$\hat{\mathcal{H}}(\boldsymbol{\theta}) = \sum_{i=0}^{N-2} \underbrace{\left(J_x \hat{X}_i \hat{X}_{i+1} + J_y \hat{Y}_i \hat{Y}_{i+1} + J_z \hat{Z}_i \hat{Z}_{i+1}\right)}_{\text{exchange of neighbours}} + \sum_{i=0}^{N-1} \underbrace{\left(h_x \hat{X}_i + h_z \hat{Z}_i\right)}_{\text{fields}} .$$

The Pauli terms are known from the design of the device. The five coefficients $\boldsymbol{\theta} = (J_x, J_y, J_z, h_x, h_z)$ are the unknowns, with true values $(1, 0.6, 0.3, 0.5, 0.2)$ in the simulation. Every coefficient multiplies a fixed matrix, $\hat{\mathcal{H}}(\boldsymbol{\theta}) = \sum_j \theta_j \hat{P}_j$, and we set $\hbar = 1$. Fitting them to time-resolved data is trajectory-based Hamiltonian learning.

C. E. Granade *et al.*, New J. Phys. **14**, 103013 (2012), doi:10.1088/1367-2630/14/10/103013 N. Wiebe *et al.*, PRL **112**, 190501 (2014), doi:10.1103/PhysRevLett.112.190501 J. Wang *et al.*, Nat. Phys. **13**, 551 (2017), doi:10.1038/nphys4074

The Learning Problem

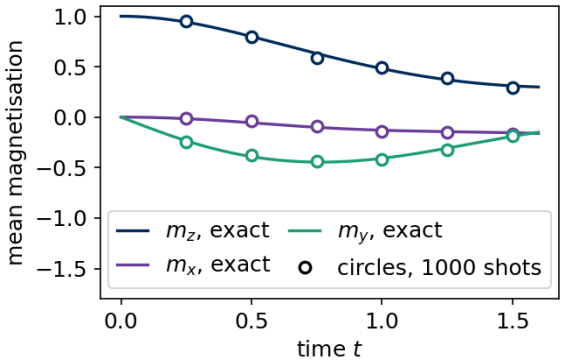


model $\hat{\mathcal{H}}(\boldsymbol{\theta}) = \sum_i \left(J_x \hat{X}_i \hat{X}_{i+1} + J_y \hat{Y}_i \hat{Y}_{i+1} + J_z \hat{Z}_i \hat{Z}_{i+1}\right) + \sum_i \left(h_x \hat{X}_i + h_z \hat{Z}_i\right)$

the Pauli terms are known, and the coefficients $\boldsymbol{\theta} = (J_x, J_y, J_z, h_x, h_z)$ are to be learned from the bit strings

Initial states the probes $|0000\rangle$, $|0101\rangle$, $|{+}{+}{+}{+}\rangle$. **Operations** evolution for $t_n = 0.25n$, $n = 1, \ldots, 6$, rotation of all qubits into the Z, X or Y basis, M shots for each of the 54 configurations. **Goal** the five coefficients. **Success** in simulation, a small relative error $\epsilon = \sum_j |\tilde{\theta}_j - \theta_j| / \sum_j |\theta_j|$.

Magnetisations from Bit Strings



our simulation

Mean magnetisations $m_g(t) = \frac{1}{N}\sum_i \langle \hat{\sigma}_i^g \rangle$ for the probe $|0000\rangle$, from 1000 shots per time (circles) and exact (lines).

The field h_x turns the spins from $+z$ towards $-y$, and the couplings dephase them. The likelihood uses the full bit strings with all correlations.

A Differentiable Model of the Experiment

A candidate $\boldsymbol{\theta}$ predicts the probability of every bit string in configuration $c = (\psi_0, g, t_n)$,

$$|\psi_{\boldsymbol{\theta}}(t)\rangle = e^{-i\hat{\mathcal{H}}(\boldsymbol{\theta})t}|\psi_0\rangle, \qquad p_{\boldsymbol{\theta}}(\mathbf{b}\,|\,c) = \Big|\langle \mathbf{b}| \underbrace{\hat{U}_g}_{\hat{I},\ \hat{H}^{\otimes N},\ (\hat{H}\hat{S}^\dagger)^{\otimes N}} |\psi_{\boldsymbol{\theta}}(t_n)\rangle\Big|^2.$$

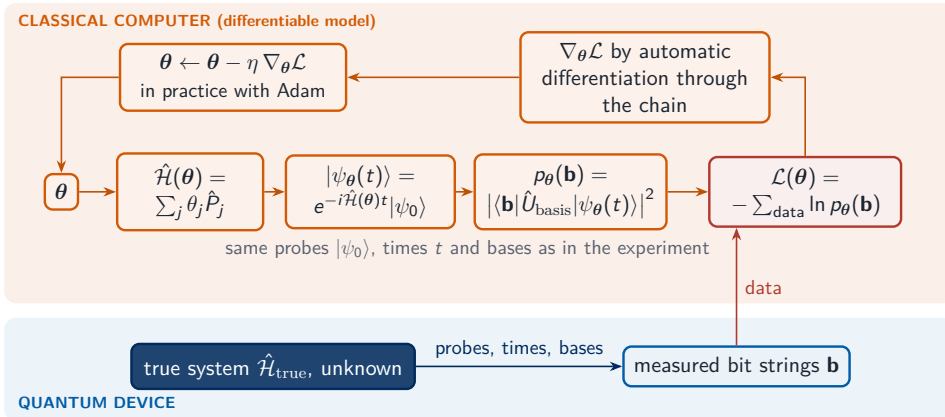
The shots are independent, and the negative log-likelihood per shot of the counts $n_c(\mathbf{b})$ is

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{M_{\mathrm{tot}}} \sum_c \sum_{\mathbf{b}} \underbrace{n_c(\mathbf{b})}_{\text{data}} \ln \underbrace{p_{\boldsymbol{\theta}}(\mathbf{b}\,|\,c)}_{\text{model}}.$$

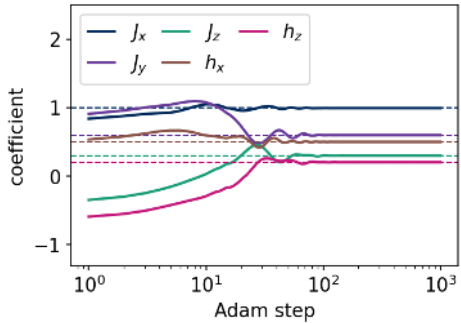
It contains only counts and model probabilities. Scaling and squaring, $e^{\hat{A}} \approx [\sum_{l=0}^{12} (\hat{A}/2^6)^l/l!]^{2^6}$, is a chain of differentiable matrix products. Gradient descent $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, \nabla_{\boldsymbol{\theta}} \mathcal{L}$ is carried out in practice with the Adam optimiser.

N. J. Higham, SIAM J. Matrix Anal. Appl. **26**, 1179 (2005), doi:10.1137/04061101X J. Bradbury *et al.*, JAX (2018), github.com/jax-ml/jax D. P. Kingma and J. Ba, arXiv:1412.6980

The Forward Model and the Training Loop



Fitting from Random Starts

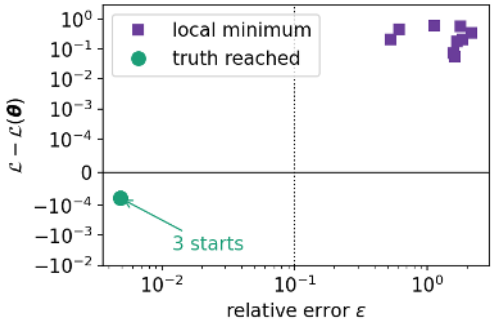


our simulation

Coefficients of the best of 12 starts drawn from $[-1, 1]^5$ during 1000 Adam steps (solid), true values dashed, $M = 1000$.

The best start reaches the true values within about 100 steps.

Local Minima



our simulation

Final loss of every start relative to the loss of the true coefficients, against its relative error ϵ.

Only 3 of 12 starts reach the truth, the others stop in local minima with a higher loss. The lowest loss selects a start with $\epsilon = 0.005$.

Real Bases Cannot Fix the Sign

The probes have real amplitudes, $|\psi_0\rangle^* = |\psi_0\rangle$. Complex conjugation of the evolved state gives

$$|\psi_{\hat{\mathcal{H}}}(t)\rangle^* = e^{+i\hat{\mathcal{H}}^* t}|\psi_0\rangle^* = e^{-i(-\hat{\mathcal{H}}^*)t}|\psi_0\rangle = |\psi_{-\hat{\mathcal{H}}^*}(t)\rangle.$$

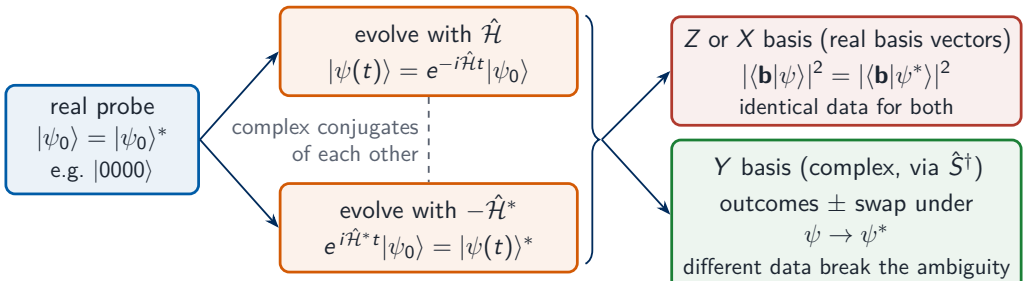
For a basis with real vectors $\langle \mathbf{b}|\hat{U}_g$ is a real row, and

$$p_{-\hat{\mathcal{H}}^*}(\mathbf{b}\,|\,c) = \left|\langle \mathbf{b}|\hat{U}_g|\psi\rangle^*\right|^2 = \left|\langle \mathbf{b}|\hat{U}_g|\psi\rangle\right|^2 = p_{\hat{\mathcal{H}}}(\mathbf{b}\,|\,c) \qquad \text{for } g = Z, X.$$

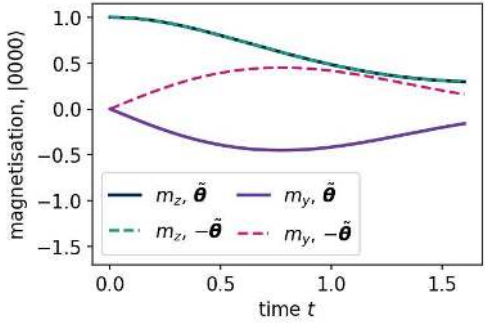
Data in the Z and X bases from real probes therefore cannot distinguish $\hat{\mathcal{H}}$ from $-\hat{\mathcal{H}}^*$ at any time and with any number of shots. The Y basis contains $\hat{S}^\dagger = \mathrm{diag}(1, -i)$, and $\hat{Y}^* = -\hat{Y}$ reverses every y magnetisation, so Y data break the tie, as does a complex probe such as $|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$.

For the XYZ chain every $\hat{P}_j$ is real, since $\hat{Y}_i\hat{Y}_{i+1}$ is real, so $-\hat{\mathcal{H}}(\boldsymbol{\theta})^* = \hat{\mathcal{H}}(-\boldsymbol{\theta})$ and the partner has all five signs flipped.

Two Hamiltonians with Identical Real-Basis Data



The Ambiguity in the Magnetisation

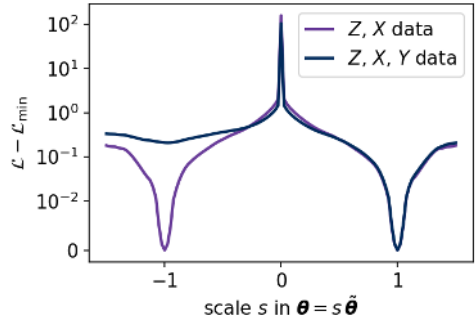


our simulation

z and y magnetisations of the probe $|0000\rangle$ under $\tilde{\boldsymbol{\theta}}$ and $-\tilde{\boldsymbol{\theta}}$.

The z magnetisation is the same for both Hamiltonians and the y magnetisation changes sign.

The Ambiguity in the Loss



our simulation

Loss along the line $\theta = s\tilde{\theta}$ with Z and X data only and with Y data added.

With real bases alone the loss has two equal minima at $s = \pm 1$. The Y data raise the loss of $-\tilde{\theta}$ by 0.63 nats per shot of Y data.

Fisher Information and the Cramér–Rao Bound

The Fisher information matrix measures how strongly the outcome probabilities respond to the coefficients. For one shot per configuration,

$$\mathcal{I}_{jl}(\boldsymbol{\theta}) = \sum_{c} \sum_{\mathbf{b}} \frac{1}{p_{\boldsymbol{\theta}}(\mathbf{b}|c)} \frac{\partial p_{\boldsymbol{\theta}}(\mathbf{b}|c)}{\partial \theta_j} \frac{\partial p_{\boldsymbol{\theta}}(\mathbf{b}|c)}{\partial \theta_l}.$$

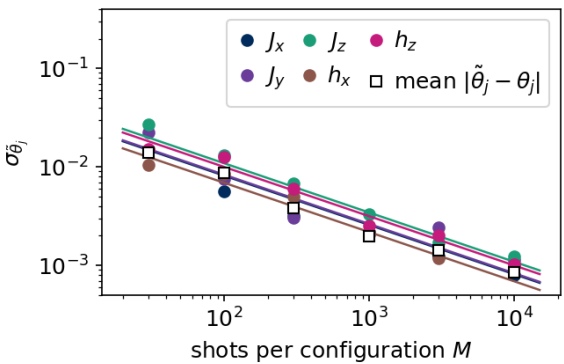
A coefficient to which no probability responds gives a zero row and cannot be learned. With M independent shots per configuration, any unbiased estimator obeys $\mathrm{Cov}(\tilde{\boldsymbol{\theta}}) \succeq \mathcal{I}^{-1}/M$, so

$$\sigma_{\tilde{\theta}_j} \;\geq\; \sqrt{(\mathcal{I}^{-1})_{jj}/M} \;\propto\; \underbrace{M^{-1/2}}_{\text{shot-noise scaling}},$$

and the maximum-likelihood estimate reaches it for many shots.

H. Cramér, *Mathematical Methods of Statistics* (Princeton, 1946), doi:10.1515/9781400883868 C. R. Rao, Bull. Calcutta Math. Soc. **37**, 81 (1945), doi:10.1007/978-1-4612-0919-5_16 S. L. Braunstein and C. M. Caves, PRL **72**, 3439 (1994), doi:10.1103/PhysRevLett.72.3439

Error Bars Reach the Bound



our simulation

Standard deviation of each coefficient over 10 independent records (markers) and the Cramér–Rao bounds (lines) against the number of shots per configuration. Squares give the mean absolute error.

The mean error falls with the slope -0.49. From $M = 300$ on the measured spread follows the bound, with a mean ratio 1.02. At $M = 1000$ the bounds range from 2.2×10^{-3} (h_x) to 3.4×10^{-3} (J_z).

The PXP Model of Rydberg Atoms

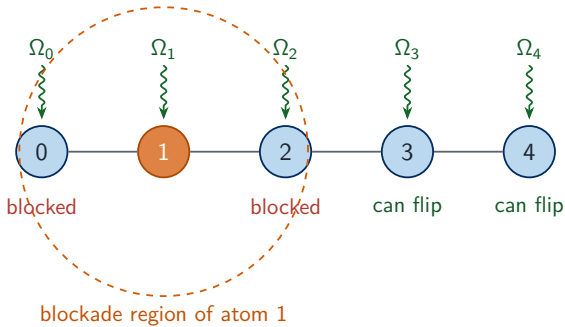
In neutral-atom arrays each atom is a qubit with ground state $|0\rangle = |g\rangle$ and Rydberg state $|1\rangle = |r\rangle$. A laser drives atom i between them at the Rabi frequency Ω_i, and neighbours cannot both be excited (Rydberg blockade). With perfect blockade,

$$\hat{\mathcal{H}}_{\mathrm{PXP}}(\boldsymbol{\Omega}) = \sum_{i=0}^{N-1} \Omega_i \underbrace{\hat{P}_{i-1}^{(0)} \hat{X}_i \hat{P}_{i+1}^{(0)}}_{\text{flip only if both neighbours are in } |0\rangle}, \qquad \hat{P}_i^{(0)} = |0\rangle\langle 0|_i,$$

with $\hat{P}_{-1}^{(0)} = \hat{P}_N^{(0)} = \hat{I}$. The unknown Ω_i are drawn from $[-1, 1]$. The probes are computational-basis states without neighbouring excitations, for four atoms $|1000\rangle$, $|0101\rangle$ and $|0010\rangle$, measured in the Z, X and Y bases at $t_n = 0.2n$. Conjugation with $\hat{Z}_i$ flips the sign of Ω_i alone and leaves Z data unchanged, so only X and Y data fix the signs.

H. Bernien *et al.*, Nature **551**, 579 (2017), doi:10.1038/nature24622 P. Fendley, K. Sengupta and S. Sachdev, PRB **69**, 075106 (2004), doi:10.1103/PhysRevB.69.075106

A Rydberg Chain



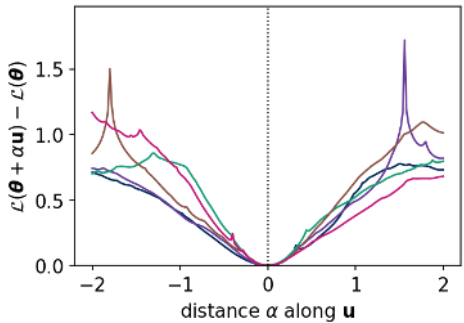
$|0\rangle = |g\rangle$, ground state

$|1\rangle = |r\rangle$, Rydberg state

a laser drives each atom with the Rabi frequency Ω_i, and the term $\hat{P}^{(0)}_{i-1}\hat{X}_i\hat{P}^{(0)}_{i+1}$ flips atom i only if both neighbours are in $|0\rangle$

C. J. Turner *et al.*, Nat. Phys. **14**, 745 (2018), doi:10.1038/s41567-018-0137-5

Loss Cuts for the XYZ Chain

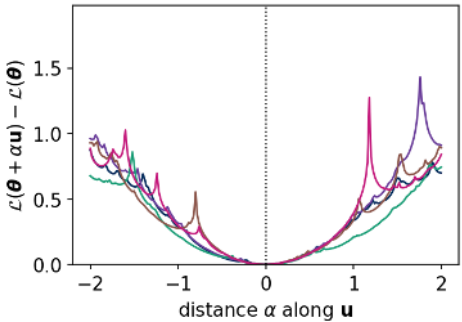


our simulation

Loss $\mathcal{L}(\boldsymbol{\theta} + \alpha\mathbf{u}) - \mathcal{L}(\boldsymbol{\theta})$ along five random unit directions $\mathbf{u}$ through the true coefficients of the XYZ chain.

The cuts contain a median of 6 local minima away from the centre.

Rugged Loss Landscapes

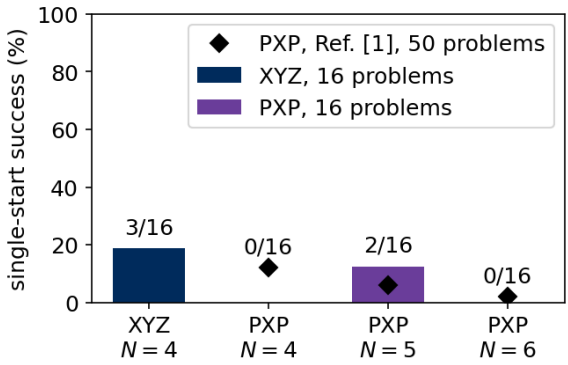


our simulation

The same cuts for the PXP model with four atoms.

The PXP cuts contain a median of 12 local minima along such cuts, against 6 for the XYZ chain.

Single Starts Fail



Fraction of 16 random problems for which a single start reaches $\epsilon < 0.1$. Diamonds are published rates of the same fixed-model fit for the PXP model, measured with different data.

Single starts rarely succeed for either model here, so our runs do not separate the two models. The contrast rests on the published rates.

T. Heightman, E. Jiang and A. Acín, Quantum Sci. Technol. **10**, 045072 (2025), doi:10.1088/2058-9565/ae0d79, Table 1 of arXiv:2408.08639v1

Neural Differential Equations

A neural ordinary differential equation has a network in its right-hand side. For the assumed Hamiltonian $\hat{\mathcal{H}}_A(\boldsymbol{\theta})$,

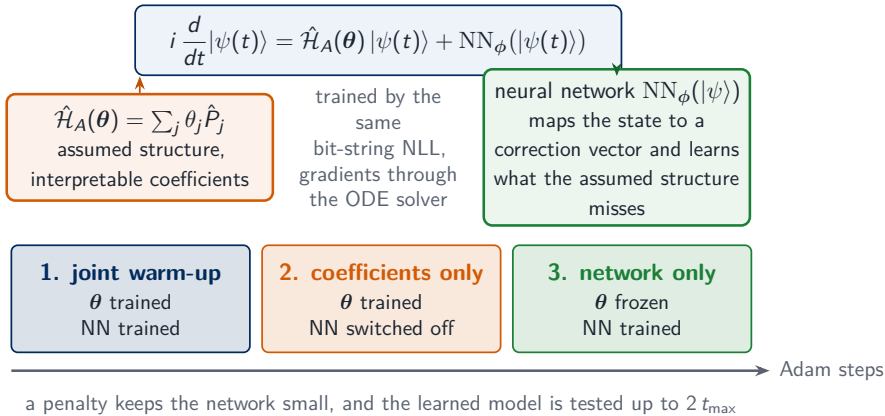
$$i\frac{d}{dt}|\psi(t)\rangle = \underbrace{\hat{\mathcal{H}}_A(\boldsymbol{\theta})|\psi(t)\rangle}_{\text{assumed form, interpretable } \boldsymbol{\theta}} + \underbrace{\mathrm{NN}_{\boldsymbol{\phi}}(|\psi(t)\rangle)}_{\text{state-dependent correction vector}},$$

where the network with parameters $\boldsymbol{\phi}$ maps the amplitudes of the state to a correction vector.

- The equation is solved numerically in small time steps, and the loss is differentiated through every step by automatic differentiation.
- The correction need not conserve the norm, so the state is normalised before the Born rule.

R. T. Q. Chen *et al.*, NeurIPS **31** (2018), arXiv:1806.07366 T. Heightman, E. Jiang and A. Acín, Quantum Sci. Technol. **10**, 045072 (2025), doi:10.1088/2058-9565/ae0d79

The Curriculum



The Network Guides the Coefficients During the Warm-Up

The plain loss is rugged in the coefficients $\boldsymbol{\theta}$, and a gradient method that starts far from the true values stops in a local minimum.

1. **Warm-up, θ and ϕ together.** The source reports that the plain fit converges only from particular starts, because of local minima and flat regions of the loss, and that the warm-up removes this dependence. The loss of the combined model is smoother.

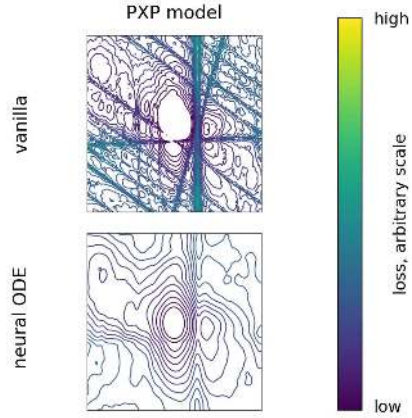
2. **Coefficients alone, network off.** The model is again the assumed Hamiltonian, so the reported coefficients are those of a Hamiltonian.

3. **Network alone, θ frozen.** The network absorbs what the assumed form cannot describe.

The published landscapes and our control, the same schedule without the network, test this picture.

T. Heightman, E. Jiang and A. Acín, Quantum Sci. Technol. **10**, 045072 (2025), doi:10.1088/2058-9565/ae0d79

Published Result: Smoother Landscapes



Loss of the PXP model in a plane spanned by two random directions through the global minimum (centre). Top, the fixed-model fit (vanilla). Bottom, the neural differential equation. Contour colour runs from dark violet (low loss) to yellow (high loss) on an arbitrary scale.

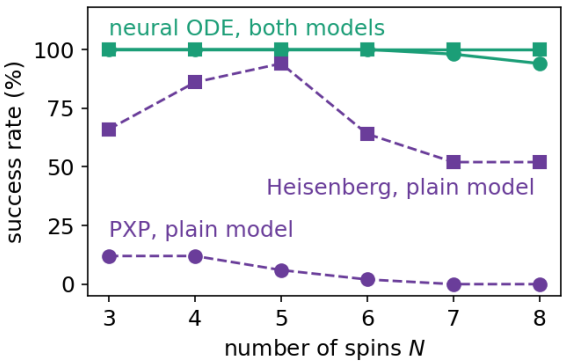
The network turns many narrow valleys into one smooth basin around the global minimum.

published result, PXP column, not our simulation

T. Heightman, E. Jiang and A. Acín, Quantum Sci. Technol. **10**, 045072 (2025), doi:10.1088/2058-9565/ae0d79,

Fig. 4 of arXiv:2408.08639v1

Published Result: Success Rates



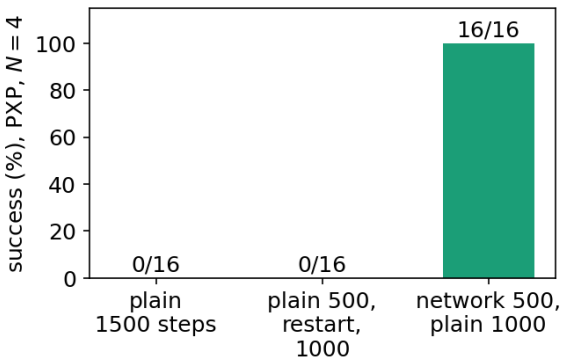
published data, replotted

Success rates of the fixed-model fit (violet, dashed) and the neural differential equation (green, solid), circles for the PXP model and squares for the Heisenberg chain in a field.

For the PXP model the fixed-model fit fails more often with growing N. The neural differential equation succeeds in almost every run up to eight spins.

T. Heightman, E. Jiang and A. Acín, Quantum Sci. Technol. **10**, 045072 (2025), doi:10.1088/2058-9565/ae0d79, Table 1 of arXiv:2408.08639v1

The Warm-Up Repairs the Fit

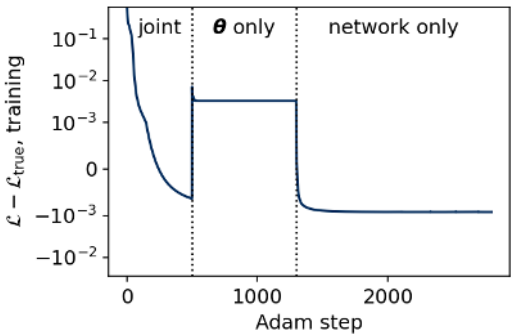


our simulation

Successful fits of the PXP model, $N = 4$, among 16 problems with the same starts. Plain model for 1500 Adam steps, a control with 500 plain steps, a restart of Adam and 1000 further steps, and a warm-up of 500 steps with the network followed by 1000 plain steps.

Only the warm-up succeeds, in all 16 problems. The network is switched off before the final fit, so the coefficients come from the plain model and the gain comes from the warm-up itself.

A Term Missing from the Model



our simulation Training loss through the three phases relative to the loss of the true Hamiltonian.

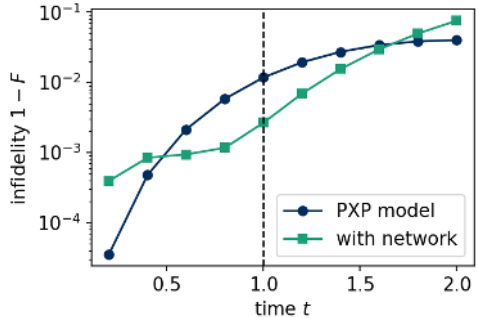
The simulated device adds a next-nearest-neighbour interaction that the PXP model lacks,

$$\hat{\mathcal{H}}_{\text{true}} = \hat{\mathcal{H}}_{\text{PXP}}(\boldsymbol{\Omega}) + V_2 \sum_i \hat{n}_i \hat{n}_{i+2},$$

with $\hat{n} = |1\rangle\langle 1|$ and $V_2 = 0.4$, and the data cover $t \leq t_{\text{max}} = 1$.

The fitted Rabi frequencies have $\epsilon = 0.013$, since the plain fit absorbs part of the missing term. Switching the network off raises the loss above the reference, and the final network-only phase lowers it slightly below, fitting part of the shot noise.

Beyond the Training Times

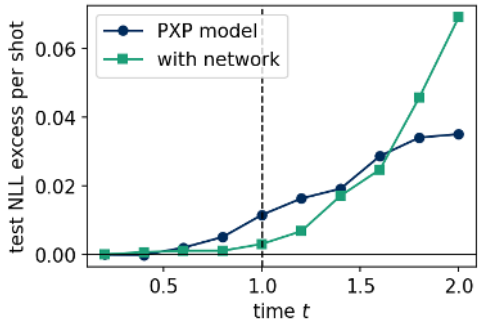


our simulation

Infidelity $1 - F(t)$ with the state of the device, mean over three probes, for the fitted PXP model with and without the network, dashed line at $t_{\max}$.

The network lowers the infidelity only between about $0.6\, t_{\max}$ and $1.6\, t_{\max}$, at $t_{\max}$ from 1.2×10^{-2} to 2.7×10^{-3}. It is slightly worse at the shortest times and worse beyond.

Test Likelihood Beyond the Training Times



our simulation

Excess of the test negative log-likelihood per shot over that of the true Hamiltonian, from fresh bit strings, which an experimenter can evaluate.

The same picture follows from data alone. At $2t_{\max}$ the network is worse than the plain model, with infidelities 7.5×10^{-2} against 4.0×10^{-2}.

10
Summary

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Summary

1. A density matrix of N qubits is fixed by the $4^N - 1$ Pauli expectation values, which rotations into three bases deliver from 3^N measurement settings.

2. Linear inversion of the Born rule is an average of parities with an error proportional to $M^{-1/2}$. It ignores positivity and is unphysical for nearly every data set of a nearly pure state.

3. Projection onto the physical states repairs the spectrum at the cost of one diagonalisation. Maximum likelihood, by the $R\rho R$ iteration or by gradient descent on a Cholesky factor, finds the most likely physical state.

4. Physical estimates are biased near the boundary of the state space. A network trained on pairs of an estimate and its true state acts as a denoising filter on the Cholesky factor of the estimate.

Summary, Continued

5. The fidelity is a maximum over amplitudes. The Cholesky factors are one pair of amplitudes, so the squared error of the Cholesky vectors bounds the squared Bures distance and serves as the loss.

6. A convolutional layer mixes neighbouring entries with weights that are fixed after training. Attention trains a rule of comparison $g = W_{\mathrm{L}}^{T} W_{\mathrm{R}}$, turns the scores into Boltzmann weights and averages the content of all vectors with them.

7. The training data fix the rule of comparison, and the present input fixes the weights. For $W_{\mathrm{L}} = W_{\mathrm{R}}$ the rule is a trained metric.

8. In the published results the network improves linear inversion and maximum likelihood at few shots, recovers the Fisher information of one-axis-twisted states it never saw in training, and outperforms networks of convolutions alone on pure states.

Summary, Concluded

9. Hamiltonian learning fits the coefficients of an assumed form to time-resolved bit strings by minimising the negative log-likelihood with gradients through a differentiable model of the experiment. Its error bars follow the Cramér–Rao bound and fall as $M^{-1/2}$.

10. Data in real bases from real probes cannot tell $\hat{\mathcal{H}}$ from $-\hat{\mathcal{H}}^*$. The Y basis or a complex probe breaks this ambiguity.

11. Single starts of the plain fit often fail, in our runs for the XYZ chain and for the PXP chain alike. A warm-up with a neural differential equation repairs the fit, and a learned correction can absorb a missing term along the training trajectories without extrapolating reliably beyond them.

11

Additional Material

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Weighted Least Squares

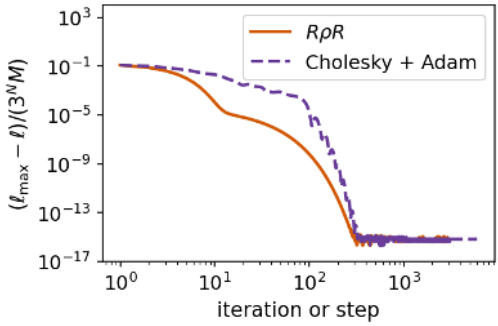
A frequency f has the variance $p(1-p)/M$, so outcomes with probabilities near 0 or 1 are more precise. Weighted least squares divides every squared residual by its estimated variance,

$$\tilde{\boldsymbol{r}}_{\mathrm{WLS}} = \arg\min_{\boldsymbol{r}} \sum_{s,\mathbf{b}} \frac{\left[(A\boldsymbol{r})_{(s,\mathbf{b})} - f_s(\mathbf{b})\right]^2}{v_s(\mathbf{b})}, \qquad v = \frac{\bar{f}(1-\bar{f})}{M}, \quad \bar{f} = \frac{n+1}{M+2},$$

where the smoothed frequency $\bar{f}$ avoids a zero variance for outcomes never observed.

For the Werner state with $M = 200$ the mean Hilbert–Schmidt distance changes only from 0.104 to 0.102, because all nine settings are measured equally often. The weighting matters when settings receive different numbers of shots. Both estimators ignore positivity.

Both Algorithms Reach the Maximum

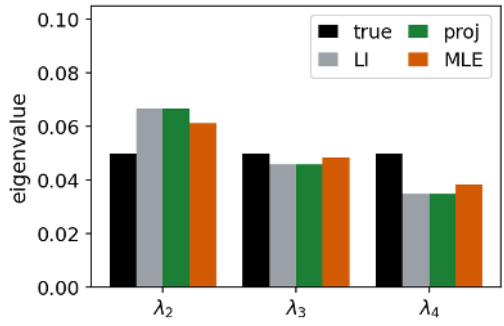


our simulation

Werner state, $M = 2000$. Distance of the log-likelihood per shot from its maximum for $R\rho R$ and for the Cholesky form with Adam.

Both reach the maximum after about 300 steps and agree to numerical precision.

Small Eigenvalues of the Estimates

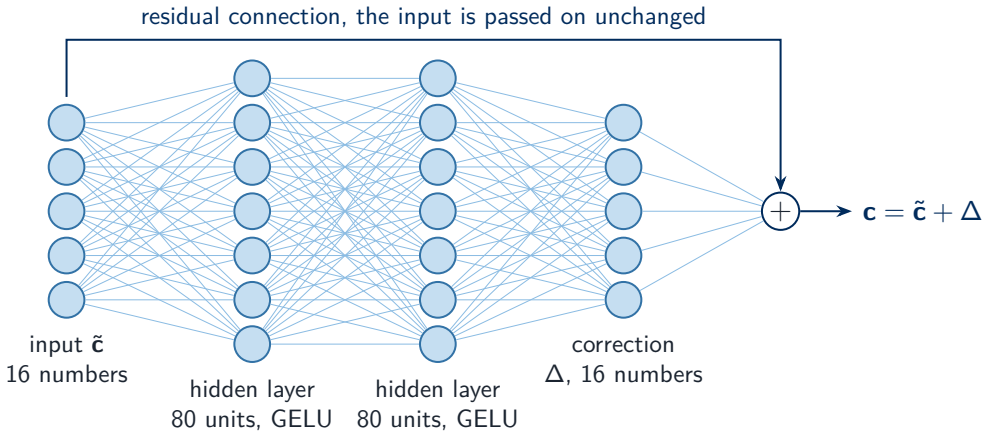


our simulation

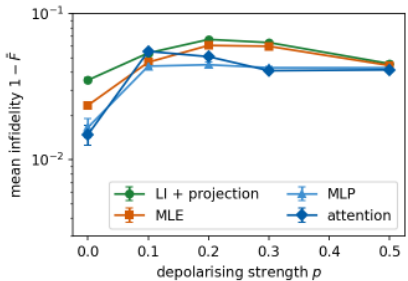
The three small eigenvalues of the true Werner state and of the three estimates from the same data.

The projected estimate has $\lambda_{\max}(\hat{R}) = 1.027 > 1$ and is no maximum of the likelihood.

The Multilayer Perceptron



Depolarised Test States



our simulation

Mean infidelity of 300 states $\hat{\rho}_p = (1-p)|\psi\rangle\langle\psi| + p\,\hat{I}/d$ with Haar-random $|\psi\rangle$, $M = 100$, networks trained on pure and Hilbert–Schmidt states only.

The spectrum of one large and three equal small eigenvalues is absent from training. The MLP still improves on the projection and on MLE at every p, the attention network at every p except 0.1.

Other Routes to a Hamiltonian

Eigenstates and steady states. A stationary state has $\langle i[\hat{\mathcal{H}}, \hat{O}_a]\rangle = 0$ for every operator $\hat{O}_a$. With $\hat{\mathcal{H}} = \sum_j \theta_j \hat{P}_j$ these conditions are linear, $B\boldsymbol{\theta} = 0$ with $B_{aj} = \langle i[\hat{P}_j, \hat{O}_a]\rangle$, and the coefficients follow from the null space of the measured matrix B up to a common factor.

Thermal states. Local expectation values of a Gibbs state $e^{-\beta\hat{\mathcal{H}}}/Z$ fix the coefficients, with sample-efficient and, at high temperature, optimal algorithms.

Short times. After a quench, $\frac{d}{dt}\langle\hat{O}\rangle = i\langle[\hat{\mathcal{H}}, \hat{O}]\rangle$ is linear in the coefficients.

E. Bairey, I. Arad and N. H. Lindner, PRL **122**, 020504 (2019), doi:10.1103/PhysRevLett.122.020504 X.-L. Qi and D. Ranard, Quantum **3**, 159 (2019), doi:10.22331/q-2019-07-08-159 A. Anshu *et al.*, Nat. Phys. **17**, 931 (2021), doi:10.1038/s41567-021-01232-0 J. Haah, R. Kothari and E. Tang, FOCS 2022, 135, doi:10.1109/FOCS54457.2022.00020 Z. Li, L. Zou and T. H. Hsieh, PRL **124**, 160502 (2020), doi:10.1103/PhysRevLett.124.160502 D. Stilck França *et al.*, Nat. Commun. **15**, 311 (2024), doi:10.1038/s41467-023-44012-5

Three Further Routes

eigenstate or steady state

$\langle i[\hat{\mathcal{H}}, \hat{O}_a] \rangle = 0$
for every $\hat{O}_a$

$$B\boldsymbol{\theta} = 0$$

$B_{aj} = \langle i[\hat{P}_j, \hat{O}_a] \rangle$
measured, $\boldsymbol{\theta}$ from
the null space of B

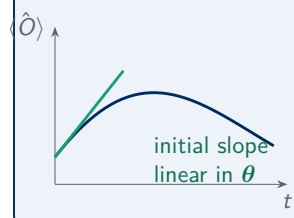
thermal state

$$\hat{\rho} = e^{-\beta \hat{\mathcal{H}}(\boldsymbol{\theta})} / Z$$

match measured local
expectation values

sample-efficient,
optimal at high temperature

short times after a quench



Learning Decay and Dephasing Rates

An open qubit obeys the Lindblad equation,

$$\frac{d\hat{\rho}}{dt} = -i[\hat{\mathcal{H}}, \hat{\rho}] + \sum_{\mu} \Big(\hat{L}_\mu \hat{\rho} \hat{L}_\mu^\dagger - \tfrac{1}{2}\{\hat{L}_\mu^\dagger \hat{L}_\mu, \hat{\rho}\} \Big), \qquad \hat{L}_1 = \underbrace{\sqrt{\Gamma}\,\hat{S}^+}_{\text{decay to } |0\rangle}, \quad \hat{L}_2 = \underbrace{\sqrt{\gamma/2}\,\hat{Z}}_{\text{dephasing}},$$

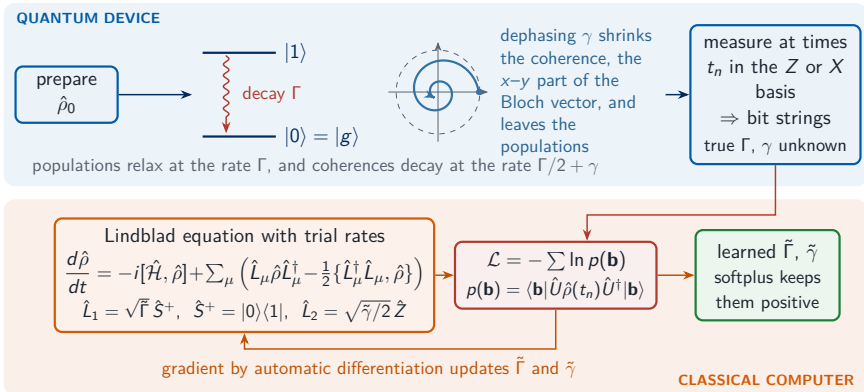
with $\hat{S}^+ = |0\rangle\langle 1|$, $|0\rangle = |g\rangle$. Without a Hamiltonian,

$$\frac{d\rho_{11}}{dt} = -\Gamma \rho_{11}, \qquad \frac{d\rho_{01}}{dt} = -\Big(\frac{\Gamma}{2} + \gamma\Big)\rho_{01}, \qquad T_1 = \frac{1}{\Gamma}, \quad \frac{1}{T_2} = \frac{\Gamma}{2} + \gamma.$$

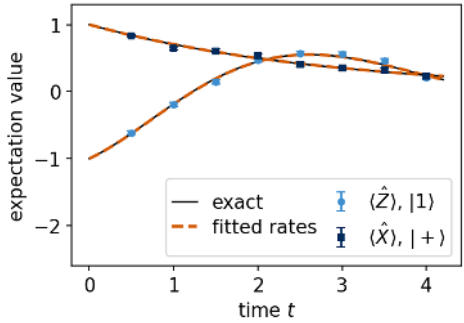
The known drive $\hat{\mathcal{H}} = (\Omega/2)\hat{X}$, $\Omega = 1$, mixes the two. **Initial states** the probes $|1\rangle$ and $|+\rangle$. **Operations** evolution for $t_n = 0.5n$, $n = 1, \ldots, 8$, and measurement in the Z or X basis. **Goal** the rates $\tilde{\Gamma}$, $\tilde{\gamma}$, kept positive through $\Gamma = \ln(1 + e^{a_\Gamma})$, $\gamma = \ln(1 + e^{a_\gamma})$, from the same negative log-likelihood. **Success** errors at the Cramér–Rao bound.

G. Lindblad, Commun. Math. Phys. **48**, 119 (1976), doi:10.1007/BF01608499 V. Gorini, A. Kossakowski and E. C. G. Sudarshan, J. Math. Phys. **17**, 821 (1976), doi:10.1063/1.522979

Rates from Bit Strings



Learned Rates

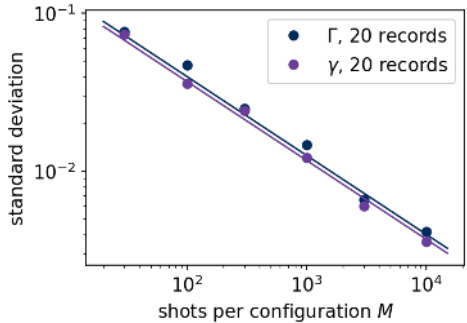


our simulation

True rates $\Gamma = 0.3$, $\gamma = 0.2$. $\langle \hat{Z} \rangle$ of the probe $|1\rangle$ and $\langle \hat{X} \rangle$ of the probe $|+\rangle$ from bit strings, exact and from the fitted model.

One record with $M = 1000$ gives $\tilde{\Gamma} = 0.297$ and $\tilde{\gamma} = 0.199$.

Error Bars of the Rates



our simulation

Standard deviation of $\tilde{\Gamma}$ and $\tilde{\gamma}$ over 20 records against M, with the Cramér–Rao bounds.

The error bars follow the bound, as for the Hamiltonian coefficients. Only the forward model changed.

Labs and Reading

Labs

- Lab 074, quantum state tomography.
- Lab 076, a neural network that denoises quantum state tomography.
- Lab 077, Hamiltonian learning.

Reviews and key papers

- D. F. V. James *et al.*, Measurement of qubits, PRA **64**, 052312 (2001), doi:10.1103/PhysRevA.64.052312.
- A. Macarone Palmieri *et al.*, Enhancing quantum state tomography via resource-efficient attention-based neural networks, Phys. Rev. Research **6**, 033248 (2024), doi:10.1103/PhysRevResearch.6.033248.
- A. Vaswani *et al.*, Attention is all you need, NeurIPS **30** (2017), arXiv:1706.03762.
- T. Heightman, E. Jiang and A. Acín, Solving the quantum many-body Hamiltonian learning problem with neural differential equations, Quantum Sci. Technol. **10**, 045072 (2025), doi:10.1088/2058-9565/ae0d79.